

Life cycle impacts of food systems under prospective scenarios

A case study on potato and cheese

Master Thesis



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By

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Preface

This thesis was conducted by Clément Aranda between 26 January 2026 and 26 June 2026 and constitutes 30 ECTS credits as part of the Master of Science programme in Industrial Engineering and Management at the Technical University of Denmark (DTU).

The thesis assumes that the reader has a basic understanding of Life Cycle Assessment (LCA) concepts and methodology.

The manuscript is structured in the format of a scientific journal article. Accordingly, the main text presents the research objectives, methodology, key results, and conclusions, while supplementary material (SM) is provided at the end of the thesis to document additional calculations, methodological details, and supporting information.

I would like to express my sincere gratitude to my supervisors, Oliver Jolliet, Matilda Nordman, and Robin Raupach, for their guidance and support throughout this work. I am also thankful to Raoul Nicolodi for his contribution to the modelling of electric tractors and to George Manthos for his input on environmental learning theory.

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1 Abstract

Food systems are a major driver of transgression of planetary boundaries, including climate change. As greenhouse gas emissions from the food sector are projected to increase by 2050, mitigation measures are required to reduce the environmental footprint of diets and ensure compliance with planetary boundaries. While dietary shifts towards more plant-based diets constitute an important mitigation strategy, transforming food production systems also appear necessary. In this study, prospective Life Cycle Assessment (pLCA) methods were applied to quantify future climate change impacts of food systems under production-side mitigation measures, using potato and cheese as case studies. Prospective changes were modeled by combining Integrated Assessment Model (IAM)-based background transformations from the REMIND-SSP1-PkBudg650 scenario with foreground projections based on either explicit mitigation measures or on environmental learning theory. Absolute Environmental Sustainability Assessment (AESA) was used to compare projected climate change impacts against product-specific sustainability targets derived from the planetary boundary framework.

Mitigation strategies affecting electricity production and road transport, notably the transition to renewable energy and the electrification of freight transport, were found to substantially reduce climate change impacts of both cheese and potato production systems by 2050. Although some burden shifting was observed in other impact categories, overall endpoint impacts decreased for both products. Additional interventions within agri-food systems further reduced climate change impacts, particularly through improvements in milk production systems and reductions in on-farm emissions from potato production. However, when foreground modeling was based on environmental learning theory, only limited additional reductions were projected, as the relatively low projected growth rates of both products resulted in modest learning effects. Overall, under the assumptions of the scenarios considered in this study, climate change impacts in 2050 were reduced by 51.2% for cheese and 59.8% for potato production relative to the reference *ecoinvent* systems. Despite these improvements, the projected reductions were insufficient to bring climate change impacts within absolute sustainability thresholds under either allocation method considered. Even under the most optimistic production-side transformation pathways, climate change impacts in 2050 exceeded the sustainability targets by a factor of 5.2 for cheese and 1.9 for potato.

These findings suggest that keeping food systems within planetary boundaries cannot be achieved through production-side measures alone and will require demand-side strategies such as dietary shifts. However, they represent only one possible future pathway and should be complemented by analyses based on alternative IAM scenarios to better capture uncertainties in achievable climate change impact reductions. Future research should extend the framework to a representative basket of food products to jointly assess production-side transformations and dietary shifts, while evaluating absolute environmental sustainability at the dietary level.

2 Introduction

2.1 Context and motivations

Food systems are responsible for a wide range of environmental impacts globally, including a significant contribution to climate change, with greenhouse gas (GHG) emissions accounting for approximately one third of global anthropogenic emissions [1]. An assessment of food system impacts against the planetary boundaries framework in the latest EAT-Lancet Commission report concludes that food systems are the dominant driver of transgression of five planetary boundaries, including land-system change, biosphere integrity, and climate change [2].

In particular, emissions from food systems are projected to increase by 2050, driven by population growth and socio-economic development including rising incomes. Under business-as-usual assumptions, agricultural greenhouse gas emissions are projected to increase substantially by mid-century, with estimates reaching up to +87% between 2010 and 2050 [3].

Achieving compatibility of the food sector with the planetary boundary for climate change by 2050 will therefore require the implementation of effective mitigation measures. Such measures include a shift toward more plant-forward dietary patterns. However, dietary change alone is generally considered insufficient to meet climate mitigation targets [4]. Additional measures targeting the production side of food systems, including productivity improvements and technological innovations, are therefore of particular importance [5].

Life Cycle Assessment (LCA) is a tool particularly relevant for quantifying the environmental impacts of food systems, as it enables the identification of potential burden-shifting between environmental impact categories while providing a holistic view of impacts along food supply chains. However, conventional LCA relies on databases that represent current technologies and supply chains, and methods from the emerging field of prospective LCA (pLCA) are therefore needed to better represent future food systems. pLCA allows the quantification of the future mitigation potential of different measures and can be used in combination with absolute environmental sustainability assessment (AESAs) to benchmark environmental performance against planetary boundaries, rather than targeting arbitrary emission reductions.

This framework is therefore applied in this study to develop decarbonization pathways based on production-side mitigation measures and to assess their compatibility with absolute sustainability targets.

Food production systems encompass a wide variety of products with different specificities, such as dairy systems, aquaculture, and crop production. Due to time constraints, this study focuses on two products: potato and cheese, while a broader generalization of the approach to other food products is discussed in Section 5.2. These products were selected because they represent contrasting production systems. Cheese is representative of animal-based food systems as a processed dairy product and is one of the most greenhouse gas-intensive commonly consumed food products, with emissions of 24 kg CO₂e per kg of product [6]. Potato is representative of crop-based systems and is among the most important staple foods globally. Indeed, in terms of global dietary importance, the potato ranks fourth after maize, wheat, and rice [7].

More details about the objectives of the study and its position within the existing literature are provided in the following subsections.

2.2 Literature review: pLCA of food systems

pLCA, according to the definition proposed by Arvidsson et al. refers to “LCA that models the product system at a future point in time relative to the time at which the study is conducted” [8]. pLCA studies therefore aim to develop an inventory model representative of the product system at a future point in time of interest. It typically requires the representation of both future foreground technologies and future background systems. While foreground modeling captures technological developments within the product system under study, background databases are adapted to reflect future socio-technical conditions [8, 9].

Prospective LCA has historically focused on emerging technologies such as nanomaterials, chemical production processes, and energy systems. As reported by Thonemann et al. (2020), only four studies had applied pLCA to future food production technologies at that time. Since then, a limited but growing number of pLCA studies have addressed food systems. Most of these studies rely on ad hoc scenario construction, in which selected parameters in the life cycle inventory are modified to represent future conditions.

For instance, Bohnes et al. developed a prospective LCA of aquaculture production by constructing scenarios in which the electricity mix was adapted to future projections, while improvements in energy efficiency and feeding strategies were also incorporated to reflect a future aquaculture system [10]. Another example is provided by Deteix et al., who assessed the prospective environmental impacts of the French diet under different scenarios. Their approach included modifications of background systems such as electricity and gas mixes, as well as foreground changes related to consumer behaviour (e.g. dietary composition) and production practices (e.g. conventional versus organic agriculture).

Such studies remain limited by their partial modification of background systems, as they do not fully capture broader structural changes in socio-economic systems that influence upstream supply chains. This limitation is addressed in studies using Integrated Assessment Model (IAM)-based pLCA frameworks. In this approach, background life cycle inventory databases are systematically aligned with predefined socio-economic scenarios derived from IAMs, i.e., models describing the interactions between society, economy, and environment. Only one IAM-based pLCA study in the food sector has been identified [11]. In this study, Marquardt et al. analyse the future environmental impacts of two meals in different geographical contexts using such IAM-based approach. Adjustments in the foreground system include, for example, increased feed efficiency in animal husbandry, while the energy system is consistently updated in line with the IAM scenario assumptions.

Instead of updating the foreground system using scenario-based assumptions, some studies rely on environmental learning curve theory, while still using IAM-based scenarios for background system adjustments [12]. This approach builds on the economic principle that costs decrease with accumulated experience and extends it to environmental impacts, assuming that environmental burdens decrease as production scales up and learning effects accumulate. Applications of this method have been explored in emerging technologies such as e-ammonia production and nanomaterial manufacturing [13, 14], as well as in the energy sector, particularly in the production of photovoltaic panels [15]. However, applications in agri-food systems remain absent.

This study builds on IAM-based prospective LCA approaches applied to agri-food systems, such as that of Marquardt et al., by integrating an absolute sustainability perspec-

tive within the planetary boundaries framework. In addition, two different prospective foreground modeling approaches are compared: a scenario-based approach, as used in studies such as Marquardt and Deteix, and a learning-based modeling framework that incorporates environmental learning dynamics in agricultural systems.

2.3 Research questions

The purpose of this study is to estimate the prospective climate change impacts of food systems under mitigation measures targeting the production side and to evaluate these impacts in relation to planetary boundaries, using potato and cheese as case studies. The following research questions are considered:

- (i) How do the climate change impacts of cheese and potato production systems evolve under prospective decarbonization pathways ?
- (ii) To what extent are the projected reductions in climate change impacts sufficient to meet climate-related absolute sustainability targets by 2050?

To answer these questions, two different prospective LCA approaches, both relying on IAM-based background scenarios, are applied. The first approach adapts the foreground inventory modeling using specific agri-food mitigation scenarios (hereafter referred to as scenario-based LCA), while the second relies on modeling environmental learning effects in the foreground system (hereafter referred to as learning-based LCA). These approaches are used to generate prospective impact estimates for the year 2050.

The relationship between the resulting impacts and absolute sustainability targets is assessed using the AESA framework. Methodological considerations and modeling assumptions are presented in the following section.

3 Method

3.1 Goal and scope definition

3.1.1 LCA framework, functional unit, and LCIA method

Two food products are considered in this study: potato and cheese. For both products, the reference flow is set to 1 kg, as the objective is to assess climate change impacts on a relative basis (per kilogram of product) rather than within the context of a complete diet. The modeling relies on generic *ecoinvent* market processes with a Rest-of-World (RoW) geography, providing a representative approximation of global average production systems and associated environmental impacts. The function of the system is to provide edible food products for human consumption. Consequently, the functional unit for each product is defined as 1 kg of product at the market gate, representing the point at which the product is supplied to downstream users. The system boundaries considered for the cheese and potato products are further detailed in Section 3.1.2.

Climate change is selected as the focus of the assessment due to its major contribution to the overall environmental impact of food products and its high relevance in prospective scenario analyses. Potential burden-shifting effects on other impact categories are explored in Section 4.2.3.

The impacts of prospective decarbonization pathways are assessed over the period 2025-2050, with 2050 selected as the reference year for climate mitigation targets and long-term sustainability pathways, and 2025 representing the baseline year of the study under present-day conditions.

Denmark is chosen as the reference country for the AESA, as the results of this study are intended to be used by DTU Sustain for further research on the environmental impacts of the Danish diet.

Activity Browser (AB, version 2.11.2) is used to perform the life cycle assessment based on the *ecoinvent* cut-off database (version 3.12) [16, 17]. The impact assessment is conducted using the Environmental Footprint (EF, version 3.1) method in order to ensure consistency with the metrics used in the absolute sustainability framework, which is further detailed in Section 3.6 [18].

3.1.2 Case study system description: cheese and potato

Both processes are taken directly from *ecoinvent* without any additional modeling assumptions. Consequently, only the life cycle stages included in the selected *ecoinvent* datasets are considered. For example, impacts associated with the use phase at the consumer level (e.g., cooking and food waste) are excluded. Further details on the selected processes and their system boundaries are provided in Figures 3.1 and 3.2.

Cheese: The process studied for cheese is the *ecoinvent* dataset “market for cheese, from cow milk, fresh, unripened (RoW)”. The system boundary encompasses upstream processes related to the production of inputs for milk production, the milk production stage itself, and all downstream processes up to the market gate. Transport from producers to consumers is included, and no losses are assumed during this stage. According to the *ecoinvent* documentation, 0.4% of the cheese supply originates as a co-product of yogurt production, while the remaining share is produced through dedicated soft cheese production, primarily representing mozzarella-type cheeses. The processing of cow milk

is largely based on a life cycle assessment of six U.S. creameries producing mozzarella cheese [19]. Emissions related to manure generated during milk production are not included within the system boundary of the cheese production process in **ecoinvent**. Indeed, the cut-off system model is used, meaning that the environmental burden associated with manure is allocated to manure consumers rather than to producers.

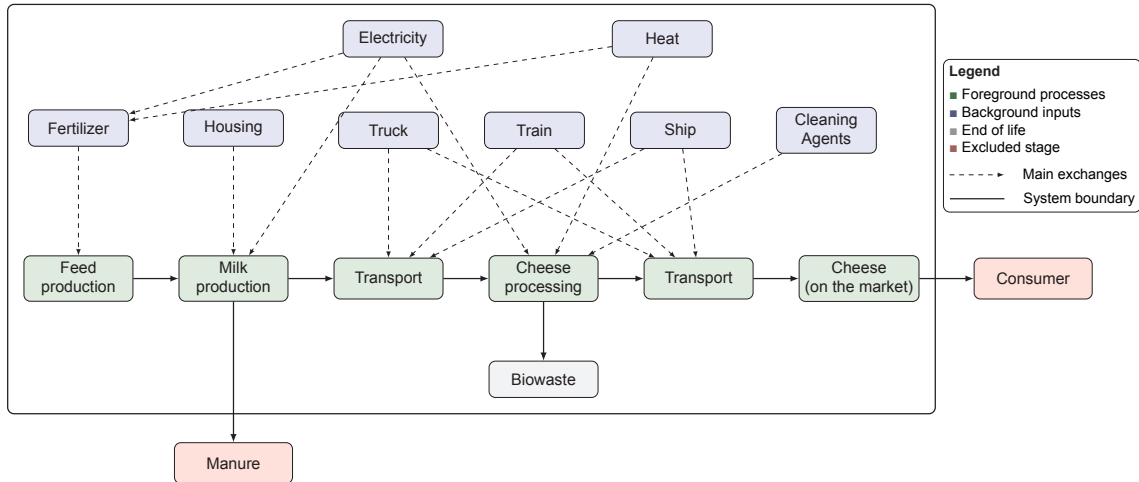


Figure 3.1: Simplified flowchart of the cheese production system including foreground, background inputs, and system boundary.

Potato: The process studied for potato is the *ecoinvent* dataset “market for potato (RoW)”, which represents a global market mix of potato production. The system boundary encompasses upstream processes related to the production of inputs for potato cultivation, the cultivation stage itself, and all downstream processes up to the market gate. Transport is included and assumed to occur under atmosphere-controlled conditions. A loss rate of 12% during transport and distribution is considered.

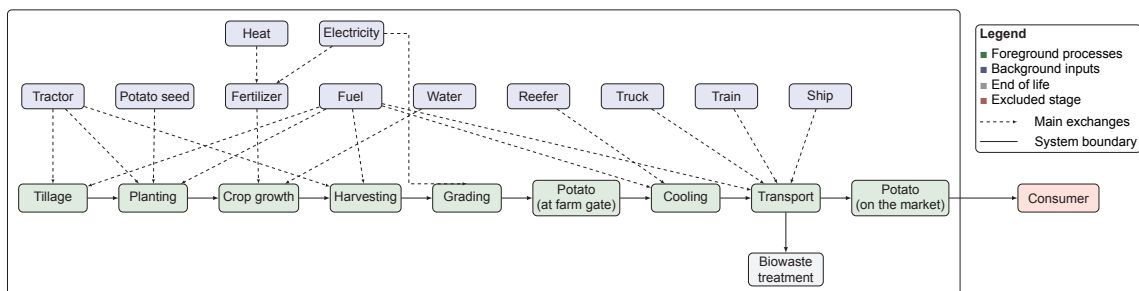


Figure 3.2: Simplified flowchart of the potato production system including foreground, background inputs, and system boundary.

3.2 Baseline definition and prospective modeling framework

3.2.1 Baseline definition and impact aggregation approach

The *ecoinvent* database is used as the baseline in this study, as it is assumed to be representative of the current environmental impacts of potato and cheese production. In the reference year 2025, all decarbonization pathways modelled in this study are assumed to have impacts equal to those of the baseline system.

Baseline impacts are first calculated at the midpoint level following the EF impact assessment method. They are then converted to endpoint results using the single-score

approach developed in EF 3.1, which aggregates all midpoint impact categories into a single indicator [20].

The endpoint impact score is computed as follows, where S_p denotes the single-score impact of product p :

$$S_p = \sum_{i \in \mathcal{I}} w_i \frac{I_{i,p}}{N_i} \quad (3.1)$$

where $I_{i,p}$ is the midpoint impact of product p in impact category i , N_i is the normalization factor for category i , and w_i is the weighting factor for category i . $\mathcal{I}$ denotes the set of all midpoint impact categories.

Normalization and weighting factors are provided in Appendix (Table 1).

In the results section, several outcomes, such as climate change hotspots, are presented as aggregated groups of activities (e.g., “Electricity production”) in order to improve the interpretability of results at the system level. Contributing activities are first grouped according to their reference product. In the *ecoinvent* database, the reference product of a process corresponds to the main good or service produced by that process (e.g., electricity for a gas-fired power plant), as illustrated in Figure 3.3.

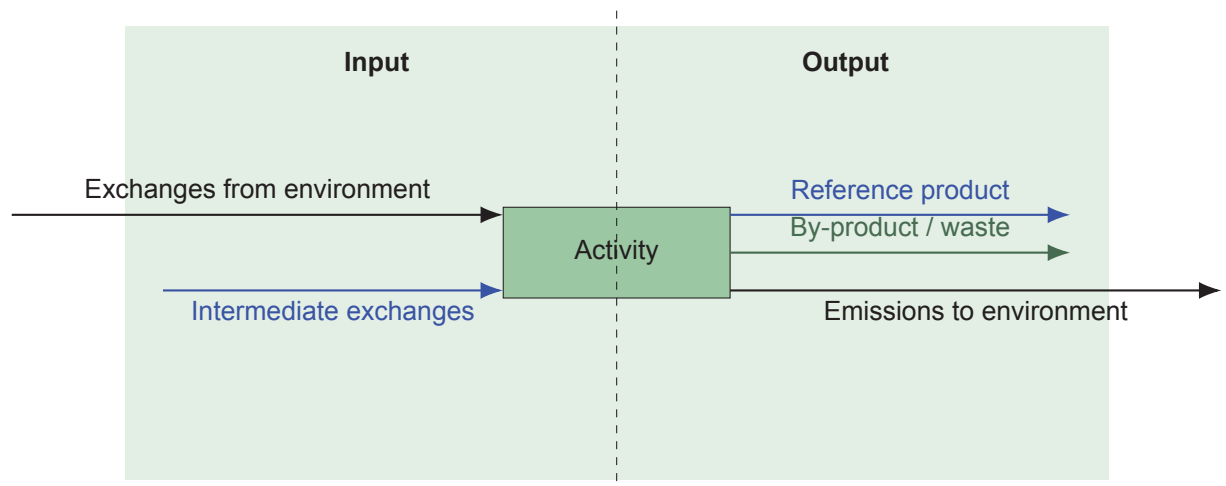


Figure 3.3: Simplified representation of an *ecoinvent* activity and associated exchanges.

These processes are then further aggregated into broader categories based on similarities in their reference products. For instance, processes producing “electricity, high voltage”, “electricity, medium voltage”, and “electricity, low voltage” are all grouped under the category “Electricity production”. It is important to note that such groupings account only for direct emissions occurring within the process itself. Upstream emissions, for example those associated with fuel extraction or processing, are not included at this aggregation level, as they are attributed to the corresponding upstream supply chain processes (e.g., fuel production).

3.2.2 Prospective modeling framework for climate change impacts

Two complementary approaches are used to estimate future climate change impacts. The first is a scenario-based approach, in which specific mitigation measures are implemented through prospective scenarios. The second is a learning-based approach relying

on environmental learning theory, where historical environmental improvement rates are extrapolated into the future.

Both approaches distinguish between background and foreground systems. Background developments are represented using socio-economic scenarios derived from IAM. These scenarios are integrated into the life cycle inventory (LCI) through prospective background databases that reflect future technological and economic conditions. In this study, IAM projections are translated into prospective *ecoinvent* databases using the Python package *premise* (version 2.3.6), which systematically modifies processes and exchanges according to the selected scenario assumptions and exports the resulting databases to Brightway for use in AB. [21]. Further details on the background modeling are given in Section 3.3.1.

Scenario-based approach

For the scenario-based approach, foreground modifications are implemented using Scenario Difference Files (SDF), following the method proposed by Steubing and de Koning [22]. An SDF is a structured Excel file that allows the modification of specific process flows in Brightway databases according to user-defined scenarios. Such a file can then be imported into AB to perform scenario-based life cycle assessments. More details on SDFs and their structure are provided in the Appendix (Table 2).

In this scenario-based approach applied in this study, foreground scenarios are constructed based on a literature review of innovative agricultural practices aimed at reducing GHG emissions. Each foreground scenario is defined for a specific target year (e.g., 2050). The corresponding SDF is then built using exchanges from a prospective background database representing the same year, thereby ensuring temporal consistency between foreground and background systems.

For a given foreground scenario, specific activities are selected within the cheese and potato production systems, such as milk production. These activities are identified through database exploration using Brightway, and the corresponding exchange information is extracted in Python to construct the SDF. Depending on the mitigation strategy considered, different technosphere or biosphere exchanges are modified. To implement these changes, additional scenario columns are introduced in the SDF, containing modified exchange values. Further details on the scenarios considered in this study and their implementation are provided in Section 3.4.

Learning-based approach

For the learning-based pLCA approach, the environmental learning theory is applied. Specific groups of foreground processes subject to learning effects are identified, and their future impacts are projected by extrapolating historical learning rates based on literature. The remainder of the system evolves according to the background scenarios. The methodology and implementation of this learning-based approach are detailed in Section 3.5.

3.3 Integration of IAM scenarios into the background system

3.3.1 Selection of IAM and scenario definition

Several IAMs are implemented within *premise*. IAMs are modeling frameworks used in climate change research to represent the interactions between society, the economy, and the environment under alternative socio-economic and climate scenarios. Each IAM can be run under multiple scenario combinations, typically defined by a Shared Socioeconomic Pathway (SSP) and a climate mitigation target. SSPs describe plausible alternative

socio-economic futures through assumptions on population growth, economic development, technological change, and policy choices. These scenarios are intended to explore a range of possible development pathways and should not be interpreted as predictions of future societal evolution.

Each IAM differs in its underlying modeling structure, with varying levels of sectoral and technological resolution. These differences translate into heterogeneous sectoral coverage and varying levels of detail in the prospective background databases generated through *premise*. Table 3.1 summarises the number of sector-specific transformations implemented by *premise* for each IAM [23].

Sector	MESSAGE	GCAM	REMIND	IMAGE	TIAM-UCL
Biomass	3	2	2	4	2
Carbon Dioxide Removal	7	4	11	5	3
Cement	26	12	4	45	7
Crops	0	10	0	10	1
Electricity	43	41	34	51	61
Fuels	24	48	49	56	55
Heat	30	13	24	38	2
Steel	14	23	42	117	119
Transport Bus	0	10	8	10	12
Transport Passenger Cars	0	30	60	10	29
Transport Rail Freight	0	10	7	10	6
Transport Road Freight	0	29	40	38	90
Transport Sea Freight	0	12	15	16	37
Transport Two Wheelers	0	6	12	0	0
Other	4	4	4	4	4
Total	151	254	312	414	428

Table 3.1: Number of sector-specific transformations implemented by *premise* for each IAM, sorted by increasing total number of transformations.

To select the most appropriate scenario for this study, two main criteria are considered. First, the selected IAM should provide a high number of transformations within *premise* and a well-balanced representation of key background sectors relevant to food systems, namely electricity, fuels, heat, and transport. Second, since the objective of this work is to assess the contribution of different decarbonization pathways towards absolute sustainability, scenarios with ambitious mitigation targets are prioritised.

Based on these criteria, the REMIND-SSP1-PkBudg650 scenario is selected (hereafter referred to as SSP1-PkBudg650). MESSAGE and GCAM are excluded due to their relatively limited number of transformations. TIAM-UCL, although providing a high level of sectoral detail overall, is excluded due to its more limited representation of the heat sector. IMAGE has the particularity of implementing modifications in the crop sector. However, these modifications are restricted to crops used for biofuel production (e.g., sugar beet and maize) and are therefore not directly relevant to the products considered in this case study. Overall, IMAGE and REMIND both provide broad and consistent coverage of the main background sectors of interest. The final selection is therefore based on the climate mitigation ambition of available scenarios. REMIND-SSP1-PkBudg650 is chosen as it exhibits the lowest projected global mean surface temperature (GMST) increase by 2100

among the available REMIND and IMAGE scenarios [24].

This scenario follows the SSP1 (“Sustainability”) pathway and assumes a cumulative CO₂ budget of 650 GtCO₂, corresponding to approximately a 50% probability of limiting warming to 2°C above pre-industrial levels, with a GMST increase of 1.17°C in 2100. It represents an optimistic mitigation pathway and is not intended to reflect current policy trajectories, but rather to explore the upper bound of potential emission reductions achievable through decarbonization of food production systems.

3.3.2 Construction of prospective life cycle inventory databases

In order to ensure temporal consistency between background and foreground modeling, SDFs describing foreground scenarios for a given year should be based on a Brightway database representative of the same year. To derive decarbonization pathways at regular time intervals over the study period (2025-2050), *premise* is applied to the *ecoinvent* 3.12 cutoff database under the SSP1-PkBudg650 scenario and run iteratively to generate prospective LCI databases for each 5-year time step between 2025 and 2050.

The main transformations implemented by *premise* for this scenario and relevant to this case study are summarized below:

- **Electricity:** Regional electricity markets are updated according to IAM regional definitions and scenario-specific generation mixes. The efficiencies of coal- and lignite-fired power plants are updated using Oberschelp et al. and further adjusted to reflect scenario-specific technological evolution [25].
- **Fuels:** Hydrogen production and distribution pathways are introduced. Regional fuel markets for diesel, gasoline, and natural gas are updated according to scenario-specific production mixes. Fuel combustion processes are adjusted to distinguish fossil and biogenic CO₂ emissions based on fuel composition.
- **Heat:** Regional heat markets are updated based on IAM regional definitions and scenario-specific technology shares. Heat production processes are relinked to updated biomass and fuel supply chains generated by *premise*.
- **Road transport:** New vehicle technologies and powertrains are introduced, including battery electric and fuel cell vehicles, based on Sacchi et al. [26]. Fleet composition is adjusted according to scenario-specific technology shares.

Two Excel files are generated to document the modifications introduced by *premise*: a scenario report and a change report. Together, these files provide a structured overview of how the LCI evolves under the prospective modeling assumptions and are used in Section 4.2.1 to analyse changes in background processes in relation to prospective results.

3.4 Development of agri-food foreground scenarios

As discussed in Section 3.3.1, *premise* only affects the transport, energy, and industrial sectors under the SSP1-PkBudg650 scenario. Additional agri-food scenarios are therefore required to assess prospective climate change impacts associated with mitigation measures implemented within the agricultural sector. The objective is to formulate scenarios targeting the main climate change hotspots of the studied products (potato and cheese) and to assess their potential to further reduce impacts and bring them closer to the planetary boundary for climate change. Such scenarios are based on technological innovations that have already been developed, for which mitigation effects have been quantified in the literature. The modeling assumptions associated with each group of agri-food scenarios investigated in this study are described in the following sections.

3.4.1 Mitigation of methane emissions from enteric fermentation

Methane emissions from dairy cows constitute a major hotspot in the climate change impact of dairy products, particularly for cheese. This has been highlighted in an LCA of mozzarella production, which is comparable to the type of cheese modelled in this study [27]. These emissions originate from enteric fermentation, a biological process in which microorganisms in the rumen break down feed during digestion in the ruminant stomach. Methane is produced as a by-product of this anaerobic digestive process and is subsequently released into the atmosphere mainly through eructation.

Several mitigation strategies have been proposed to reduce these emissions. In this study, their effects are parameterized using the meta-analysis by Arndt et al., which synthesises evidence from 430 studies evaluating 98 mitigation interventions aimed at reducing methane emissions from ruminants [28]. The impacts of these mitigation measures are reported for both emissions and animal performance metrics. Emission performance metrics include the ability to reduce methane emissions in both absolute terms (daily CH_4 emissions) and intensity terms (CH_4 emissions per kg of milk, hereafter referred to as CH_4I_m). The impacts of the mitigation measures on feed intake, fibre digestibility, and milk yield are also quantified.

To select the mitigation measures implemented in this study, CH_4I_m is used as the main comparative indicator, as the focus of the analysis is on the emission intensity of cheese and the functional unit is expressed per kilogram of product.

Three mitigation measures with the highest potential for reducing CH_4I_m are selected, and their effects are summarized in Table 3.2.

Mitigation measure	Mechanism of action	CH_4 intensity reduction
Increased feeding level	Higher dry matter intake increases animal productivity and improves feed conversion efficiency, thereby reducing methane emissions per unit of milk produced.	−16.7%
3-NOP supplementation as methane inhibitor	Reduction of methane formation in the rumen through direct inhibition of the biological processes responsible for methane production.	−31.1%
Nitrate supplementation as hydrogen sink	Redirection of hydrogen produced during rumen fermentation towards nitrate reduction, thereby acting as an alternative hydrogen sink that reduces methane formation.	−14.7%

Table 3.2: Mitigation measures targeting enteric methane emissions and associated effects on methane intensity

Then, each mitigation option is translated into a scenario characterised by specific variations in parameters reported in Arndt et al: CH_4I_m (kg CH_4 /kg milk), milk yield (kg milk/day), dry matter intake (DMI, kg/day), and methane intensity (kg CH_4 /kg milk).

Fiber digestibility has implications for manure-related emissions, as increased digestibility reduces the amount of manure produced. However, this effect is not considered in the present modeling framework, since manure emissions in the *ecoinvent* cut-off database are not attributed to milk production processes but to manure consumers.

The reduction in CH_4 intensity is not directly expressed in terms of the functional unit used

in *ecoinvent* (kg FPCM, fat- and protein-corrected milk). However, it is assumed that the considered mitigation strategies have limited effects on milk composition (fat, protein, and energy content), such that the same relative reduction can be applied to milk production processes in *ecoinvent*.

For 3-NOP and nitrate supplementation, parameter values are derived from more specific studies focusing on dose-response relationships. Indeed, these mitigation strategies rely on feed supplementation, where methane reduction depends on the administered dose, a relationship not fully captured in Arndt et al.

- **3-NOP supplementation:** The dose-response relationship is taken from a meta-analysis by Kebreab et al., where an average dose of 70.5 mg/kg DMI results in a 32.6% reduction in CH₄ intensity [29]. Effects on milk yield and DMI are assumed to be negligible, consistent with the confidence intervals reported by Arndt et al. [28].
- **Nitrate supplementation:** The dose-response relationship is taken from a meta-analysis by Dicks et al., where an average dose of 13.8 g/kg DMI leads to a 20.1% reduction in CH₄ intensity, along with associated changes in milk yield and DMI at this supplementation level [30].

The resulting parameterization is reported in Table 3.3.

Scenario	Label	Parameter	Unit	Value	Reference
Increased feeding level	FL	Yield variation	%	+16.80	Arndt et al. (2022)
		DMI variation	%	+58.30	Arndt et al. (2022)
		CH ₄ intensity variation	%	-16.70	Arndt et al. (2022)
3-NOP supplementation	NOP	Dose	kg	7.05×10^{-5}	Kebreab et al. (2023)
		Yield variation	%	0.00	Arndt et al. (2022)
		DMI variation	%	0.00	Arndt et al. (2022)
		CH ₄ intensity variation	%	-32.60	Kebreab et al. (2023)
Nitrate supplementation	NS	Dose	kg	1.38×10^{-2}	Dicks et al. (2026)
		Yield variation	%	-0.31	Dicks et al. (2026)
		DMI variation	%	-4.19	Dicks et al. (2026)
		CH ₄ intensity variation	%	-20.10	Dicks et al. (2026)

Table 3.3: Parameter values for enteric fermentation scenarios, including scenario labels and corresponding literature sources.

Scenario implementation

A SDF is generated by modifying the *ecoinvent* processes according to the scenario parameterization. This SDF affects all exchanges between milk production activities and their associated inputs. For a given milk production activity, the affected exchanges are modified as follows:

$$Q_{\text{feed,input}}^s = Q_{\text{feed,input}}^b \cdot \frac{1 + \Delta \text{DMI}}{1 + \Delta Y} \quad (3.2)$$

$$Q_{\text{CH}_4}^s = Q_{\text{CH}_4}^b \cdot (1 + \Delta \text{CH}_4) \quad (3.3)$$

$$Q_{\text{other}}^s = \frac{Q_{\text{other}}^b}{1 + \Delta Y} \quad (3.4)$$

where $Q_{\text{feed,input}}$ denotes the amount of feed input exchanges associated with milk production, and Q_{CH_4} represents methane emissions to the environment. ΔDMI denotes the relative variation in dry matter intake, ΔY the relative variation in milk yield, ΔCH_4 the relative variation in methane emissions. The superscripts b and s refer to the baseline and scenario-modified systems, respectively, where b corresponds to the original *ecoinvent* values and s to the values adjusted for scenario s .

Feed inputs are identified manually within the *ecoinvent* database, and the full list is provided in Appendix (Table 3).

The yield-scaling Equation 3.4 is applied to all technosphere and biosphere flows, except for feed inputs, methane emissions (which are already adjusted in Equation 3.2 and 3.3), and electricity consumption in milk processing. Electricity consumption in milk processing is excluded because this process is not affected by variations in yield, as it depends on the total quantity of milk processed, which remains constant since all scenarios are normalised to the functional unit.

In addition to modifications in milk production processes, new processes are introduced to account for the upstream climate change impacts of 3-NOP and nitrate production. Two dedicated activities are created in AB. Each activity includes a single biosphere flow (“Carbon dioxide, fossil”) representing its cradle-to-gate carbon footprint.

An emission factor of 52 kg CO₂e/kg 3-NOP is used for 3-NOP production, based on Feng et al. [31] (personal communication from DSM Nutritional Products). Nitrate is assumed to be supplemented to cows as calcium nitrate Ca(NO₃)₂, with an emission factor of 0.67 kg CO₂e/kg Ca(NO₃)₂, corresponding to calcium nitrate production in Europe [31, 32].

These upstream emissions depend on the quantity of product supplemented and therefore on the supplementation dose, expressed per kg DMI. Dry matter intake is first computed for each milk production activity. Feed inputs are identified and converted into dry matter using conversion factors from *ecoinvent*. Conversion factors can be found in Appendix (Table 3). The quantity of a given additive in a given milk production location l is then computed as follows:

$$Q_{\text{additive}}^l = d_{\text{additive}} \cdot \text{DMI}^l \quad (3.5)$$

where d_{additive} is the additive dose (kg additive per kg DMI), and DMI^l is the dry matter intake associated with the milk production activity in location l .

3.4.2 Mitigation of direct agricultural emissions

Direct agricultural emissions, defined as GHG emissions released in the field during crop growth, represent one of the main hotspots of climate change impacts in potato production [33]. Similarly, emissions associated with feed crop cultivation constitute a major contributor to the carbon footprint of cheese production [27].

Direct GHG emissions from agricultural systems mainly originate from biogeochemical processes occurring in soils. Nitrous oxide (N₂O) emissions are primarily generated through nitrification and denitrification, two microbial processes involved in the transformation of nitrogen following fertilizer application. Carbon dioxide (CO₂) emissions mainly arise from the decomposition of soil organic matter by microorganisms and from soil disturbance induced by cultivation practices, which accelerates the oxidation of organic carbon and its release to the atmosphere. These processes are strongly influenced by soil characteristics, climatic conditions, and agricultural management practices.

In order to formulate scenarios representative of mitigation measures, the same general approach as for modeling enteric fermentation mitigation measures is applied. Meta-analyses are first used to identify the mitigation potential of relevant agronomic practices, followed by more specific literature to refine parameter values for scenario development. Due to the limited availability of literature comparing mitigation measures specific to potato crops, mitigation measures derived from general agroecosystems (e.g., cereals and legumes) are used instead.

The first key source is a synthesis of 113 meta-analyses assessing the effects of agronomic mitigation measures on crop yield, greenhouse gas emissions (N_2O , CO_2), and soil properties [34]. Variations in crop yield and GHG emissions are of particular interest, as they directly influence climate change impacts and can be integrated into *ecoinvent* processes without explicitly modeling nutrient cycling or soil dynamics, which is beyond the scope of this thesis.

Based on Young et al., mitigation measures showing the highest reductions in GHG emissions, while maintaining neutral or positive effects on yield on average and exhibiting moderate variability across studies, are selected. The selected measures include enhanced-efficiency fertilizers and optimization of fertilizer placement.

In addition, a meta-analysis focusing specifically on the mitigation potential of N_2O emissions from agronomic practices is used to complement this selection [35]. This study highlights the effectiveness of optimizing fertilizer application rates and also provides detailed insights into enhanced-efficiency fertilizers, including nitrification inhibitors, urease inhibitors, and double inhibitors (combining nitrification and urease inhibition).

Table 3.4 summarizes the selected mitigation measures together with their mechanisms of action for reducing emissions.

Mitigation measure	Mechanism of action
Optimized fertilizer rate	Fertilizer inputs are adjusted to crop demand, reducing nitrogen surplus in soils
Optimized fertilizer placement	Improved nutrient use efficiency through closer placement of nutrients to plant roots
Use of nitrification inhibitors	Slower conversion of nitrogen in soils, reducing nitrate formation and associated emissions
Use of urease inhibitors	Slower hydrolysis of urea-based fertilizers into ammonium, reducing early nitrogen losses
Use of double inhibitors	Combined inhibition of nitrogen transformation pathways, improving nitrogen retention in soils

Table 3.4: Mitigation measures targeting direct agricultural emissions and their underlying mechanisms of action.

For each mitigation measure, the main variables of interest are the variation in crop yield, N_2O emissions, and CO_2 emissions (when available). Specific literature is then used to derive parameter values for each mitigation measure.

- **Optimized fertilizer rate**

A meta-analysis by Yao et al. covering wheat, maize, and rice systems, is used to quantify the relationships between fertilizer reduction, crop yield, and N_2O emissions [36]. This study is particularly relevant as it provides a dose-response rela-

tionship between reductions in nitrogen fertilizer inputs and the resulting changes in N₂O emissions. This enables a consistent modeling of mitigation effects that simultaneously affect field emissions and upstream emissions associated with fertilizer production. For wheat systems, an average 39.8% reduction in N₂O emissions is reported across 333 studies, together with a breakdown of results across different levels of nitrogen fertilizer reduction. However, the corresponding average reduction in fertilizer application rate associated with this effect is not explicitly reported in the study. Therefore, an arbitrary value of 50% nitrogen fertilizer reduction is assumed. This represents a conservative assumption, since the reported results indicate that studies with fertilizer reductions above 50% tend to exhibit N₂O reductions higher than the overall average of 39.8%. Yield variations are applied under the same assumptions and based on the same study. No change in CO₂ emissions is considered due to lack of available data.

- **Optimized fertilizer placement**

Parameter values are based on Wu et al., a meta-analysis of 99 studies on deep fertilizer placement [37]. This study provides estimates for yield and N₂O emission changes. No information is available regarding CO₂ emissions; therefore, no changes are assumed for this impact category.

- **Use of enhanced-efficiency fertilizers**

Yield, N₂O, and CO₂ variations are taken from Chen et al., a systematic review of environmental benefits of enhanced-efficiency fertilizers [38].

The resulting parameterization is reported in Table 3.5.

Scenario	Label	Parameter	Unit	Value	Reference
Optimized fertilizer rate	FR	N fertilizer variation	%	-50.00	Yao (2024)
		Yield variation	%	-10.90	Yao (2024)
		N ₂ O emissions variation	%	-39.20	Yao (2024)
Optimized fertilizer placement	FP	Yield variation	%	+13.50	Wu (2024)
		N ₂ O emissions variation	%	-16.20	Wu (2024)
Use of nitrification inhibitors	NI	Yield variation	%	+12.90	Chen (2023)
		N ₂ O emissions variation	%	-44.00	Chen (2023)
		CO ₂ emissions variation	%	-8.90	Chen (2023)
Use of urease inhibitors	UI	Yield variation	%	+6.20	Chen (2023)
		N ₂ O emissions variation	%	-24.00	Chen (2023)
Use of double inhibitors	NI+UI	Yield variation	%	+4.90	Chen (2023)
		N ₂ O emissions variation	%	-36.00	Chen (2023)

Table 3.5: Parameter values for direct agricultural scenarios, including scenario labels and corresponding literature sources.

Scenario implementation

A SDF is generated by modifying *ecoinvent* processes according to the scenario parameterization. This SDF affects all exchanges between potato production activities and their associated inputs. For a given potato production activity, the affected exchanges are modified as follows:

$$Q_{N_fertilizer}^s = Q_{N_fertilizer}^b \cdot (1 + \Delta N) \quad (3.6)$$

$$Q_{CO_2}^s = Q_{CO_2}^b \cdot \frac{1 + \Delta CO_2}{1 + \Delta Y} \quad (3.7)$$

$$Q_{N_2O}^s = Q_{N_2O}^b \cdot \frac{1 + \Delta N_2O}{1 + \Delta Y} \quad (3.8)$$

$$Q_{other}^s = \frac{Q_{other}^b}{1 + \Delta Y} \quad (3.9)$$

where $Q_{N_fertilizer}$ represents nitrogen fertilizer inputs and associated flows (e.g., packaging), Q_{CO_2} and Q_{N_2O} represent emissions to the environment. ΔN denotes the relative variation in nitrogen fertilizer inputs, ΔY the relative variation in crop yield, ΔCO_2 the relative variation in carbon dioxide emissions, ΔN_2O the relative variation in nitrous oxide emissions, b the baseline, and s the scenario.

$Q_{N_fertilizer}$ includes all nitrogen fertilizer inputs as well as processes that scale with nitrogen fertilizer quantity, such as fertilizer packaging, whose quantities are reduced when less fertilizer is applied. Inorganic fertilizer application (spreading) processes are not scaled, as their functional unit in *ecoinvent* is defined per surface area (ha) and is therefore not affected by changes in fertilizer quantity. However, manure spreading activities are expressed per volume and are scaled accordingly with the reduction in nitrogen inputs.

All remaining processes are scaled with yield according to Equation 3.9, except for harvesting and grading operations, which take place post-harvest and therefore depend on the quantity of potatoes harvested. This quantity remains constant across all scenarios, as results are normalised to the functional unit (1 kg of potato).

Except for the FR scenario, fertilizer quantities are not affected by the mitigation practices, only potato production processes are modified, while upstream processes remain unchanged. Enhanced-efficiency fertilizers differ from conventional fertilizers in that they require additional manufacturing steps, such as coating granules or synthesizing and incorporating nitrification inhibitors, thereby increasing production complexity compared to standard ammonia-based fertilizers. This may lead to additional upstream environmental burdens compared to conventional fertilizers. However, these effects are not considered in this study due to the limited availability of literature on this topic.

Adaptation for cow feed crops

Since direct emissions from feed crop cultivation represent a hotspot in the cheese system, the same scenarios are applied to feed crops used for cattle feeding, using the same parameterization as for potato crops (Table 3.5).

The main difference lies in the SDF generation. Exchanges between feed crop activities and their inputs are identified by exploring the milk supply chain graph in Brightway using the following procedure :

1. Identify milk production activities.
2. Retrieve technosphere inputs.
3. If an input corresponds to a feed activity, add it to the set of activities to be explored.
4. If the activity corresponds to a feed crop activity, include it in the SDF.

5. Iterate over the set of activities to be explored until no new feed-related activities are identified.

The same equations are then applied for parameterization. However, yield scaling is applied to all remaining processes without exclusion, unlike in the potato case, due to the large number of exchanges (over 26,000 modified flows in the feed crop SDF), which makes manual identification of processes that should not be scaled with yield more difficult.

3.4.3 Mitigation of agricultural machinery emissions

Apart from direct agricultural emissions, fuel combustion in agricultural machinery (i.e., associated with ploughing, planting, and harvesting operations) represents a significant hotspot of climate change impacts in agriculture [39]. Although part of these emissions is already indirectly addressed through the inclusion of hydrogen-based synthetic diesel in the fuel mix of the SSP1-PkBudg650 scenario, an additional mitigation pathway is explored through the adoption of agricultural electric vehicles. These vehicles are gaining increasing attention as part of the broader electrification of the transport sector, with a projected annual growth rate of 12% over the coming decade [40].

A scenario consisting of a switch from diesel-powered to electric agricultural machinery is therefore modelled for all activities involving tractor use. Due to time constraints, only the use phase is affected in this scenario, while tractor production and composition remain unchanged.

A scenario difference file is generated by modifying *ecoinvent* processes to represent this fuel switch. The SDF targets activities involving tractor use and affects all exchanges related to diesel inputs and biosphere resulting from fuel combustion. These activities are identified in Brightway by selecting all processes that include the input "market for tractor, 4-wheel, agricultural".

For a given activity, the emissions to the environment related to diesel combustion and the diesel input are affected as follows:

$$Q_{\text{combustion_flows}}^{ET} = 0 \quad (3.10)$$

$$Q_{\text{diesel_inputs}}^{ET} = 0 \quad (3.11)$$

Combustion flows correspond to emissions that are removed when switching from diesel combustion to electricity. Due to limited information on the origin of these elementary flows within *ecoinvent* activities, they are identified manually, and the removed flows are listed in Table 3.6.

To account for electrification, diesel-related technosphere flows are replaced by electricity inputs. Electricity is linked to the corresponding country-specific electricity mix through the activity "electricity, low voltage". Electricity demand is derived from baseline diesel consumption $Q_{\text{diesel_inputs}}^b$ and scaled using the relative efficiencies of diesel and electric powertrains:

$$Q_{\text{electricity_inputs}}^{ET} = Q_{\text{diesel_inputs}}^b \cdot \frac{\text{LHV}_{\text{diesel}}}{3.6} \cdot \frac{\eta_{\text{diesel}}}{\eta_{\text{electric}}} \quad (3.12)$$

where $\text{LHV}_{\text{diesel}} = 42.8 \text{ MJ kg}^{-1}$ is the lower heating value of diesel (based on "diesel, low-sulfur" activity in *ecoinvent*), 3.6 MJ kWh^{-1} is the conversion factor between MJ and

Elementary flow	Description
Ammonia	Emissions from engine exhaust and after-treatment.
Benzene	Unburned fuel components released during combustion.
Benzo(a)pyrene	Pollutants formed during incomplete combustion.
Fossil CO ₂	Carbon dioxide from diesel combustion.
CO (fossil)	Product of incomplete combustion.
N ₂ O	Nitrous oxide formed during combustion.
Dioxins	Trace toxic compounds from combustion processes.
Waste heat	Heat released by the engine.
CH ₄ (fossil)	Unburned methane in exhaust gases.
NO _x	Nitrogen oxides formed at high temperatures.
NM VOC	Non-methane volatile organic compounds from combustion.
PAH	Polycyclic aromatic hydrocarbons from incomplete combustion.
PM	Particulate matter emissions from diesel exhaust and resuspension.
SO ₂	Sulfur dioxide from sulfur in diesel fuel.

Table 3.6: Combustion-related elementary flows removed under the electrification scenario.

kWh, η_{diesel} is the brake thermal efficiency of the diesel tractor (35%, estimated for medium engine load, based on Tang et al. [41]), η_{electric} is the electric drivetrain efficiency (85%, based on Götz et al. [42]).

3.4.4 Integrated decarbonization pathways from IAM and scenario combinations

To formulate decarbonization pathways based on the agri-food scenarios considered, SDFs are created for each year between 2030 and 2050 at five-year intervals and for each scenario $s \in \mathcal{S}$, where the scenario space is defined as:

$$\mathcal{S} = ENT \cup AD \cup AM, \quad (3.13)$$

with ENT, AD, and AM corresponding to the scenario groups described in this section: enteric fermentation mitigation (ENT), agricultural direct emissions mitigation (AD), and agricultural machinery emissions mitigation (AM).

Each scenario s is generated based on the background database corresponding to the considered year. Then, to quantify the integrated prospective climate change impacts resulting from the combined evolution of the background system and foreground interventions under a given scenario s , a scenario LCA is performed in AB using the potato and cheese product systems from the prospective database of year t with the application of the corresponding SDF for scenario s .

In order to model the effect of implementing several mitigation measures in parallel, agri-food scenarios are combined. This is done directly in AB, where SDFs can be combined within the interface to represent joint scenarios such as 3-NOP supplementation and electrified agricultural machinery. In practice, scenarios belonging to different groups (ENT, AD, AM) can generally be combined because they act on distinct parts of the production system. For example, enteric fermentation mitigation scenarios can be combined with agricultural operations mitigation scenarios, as they affect different processes within the life cycle, namely milk production activities and feed crop production.

In contrast, combining two scenarios within the same group is generally not possible without a more detailed assessment of their interactions. For instance, all scenarios in the AD group affect N₂O emissions to the environment. Combining multiple mitigation measures would require first assessing their combined effect on this emission flow relative to a baseline situation, in order to quantify a single aggregated modification of the N₂O flux for the combined scenario.

Therefore, prospective climate change impacts are determined by combining scenarios only when this is possible. The resulting evaluated cases correspond to:

$$t \in \{2030, 2035, 2040, 2045, 2050\}, \quad s \in ENT \times AD \times AM. \quad (3.14)$$

All corresponding SDFs are provided in the Supplementary Materials (SM1).

3.5 Projection of climate change impacts based on historical learning trends

As the mitigation measures that will be implemented in the future are not yet known and their potential remains uncertain, the scenario-based approach can be complemented by the application of environmental learning theory. This theory enables the projection of future climate change impacts of food products without specifying the exact mitigation measures required, by instead relying on historical trends in environmental performance. The main modeling assumptions underlying this approach are described in the following section.

3.5.1 Estimation of historical environmental learning rates

Historical learning effects are estimated based on empirical data from the Food and Agriculture Organization of the United Nations (FAO), covering the period from 1961 to 2023 [43]. The dataset provides emissions intensity (kg CO₂e/kg commodity) together with annual production quantities.

The objective is to estimate learning effects occurring in key processes of cheese and potato production systems in order to derive prospective climate change impacts based on a projection of historical learning, leading to potential impact reductions over time. Therefore, the FAO dataset for “Raw milk of cattle” is used to represent learning in milk production systems. Since no direct potato-specific data are available, the commodity “Cereals excluding rice” is used as a proxy for potato production.

The emission intensity in these datasets represents only on-farm emissions. For milk, this includes methane emissions from enteric fermentation and emissions from manure management. For cereals excluding rice, emissions include N₂O emissions from crop residues and synthetic fertilizer application across barley, maize, millet, oats, rye, sorghum, and wheat systems.

To estimate environmental learning rates, the framework proposed by van Nielen et al. is used [12]. This approach is grounded in Wright’s law, which states that unit costs decrease as a power function of cumulative production. When translated to environmental impacts, this relationship can be expressed as:

$$CC(P) = CC_0 \left(\frac{P}{P_0} \right)^{-\alpha} \quad (3.15)$$

where CC is the emission intensity (kg CO₂e/kg product), P is cumulative production, P_0 is the reference cumulative production, and CC_0 and α are parameters representing the baseline intensity and the learning exponent, respectively.

Although the FAO dataset spans the period from 1961 to 2023, the empirical learning model is estimated only over the period 1990–2023. This choice is motivated by the fact that emission intensity dynamics prior to 1990 do not follow a consistent Wright-type learning pattern and are instead dominated by structural transitions in agricultural systems. Including this early period would therefore bias the estimation of the learning exponent α .

This relationship is fitted to empirical data over the period 1990–2023 to estimate α . The environmental learning rate (LR) is then derived as:

$$LR = 1 - 2^{-\alpha} \quad (3.16)$$

which represents the fractional reduction in environmental impact associated with each doubling of cumulative production. It represents an aggregate of all learning mechanisms correlated with cumulative production in the past, such as learning-by-doing (process optimisation through experience) and production-driven technological improvements. Such learning effects occurring in the milk and potato production systems are projected onto the same emission boundary, meaning that learning is assumed in the future only for the on-farm emissions from milk and potato production.

For milk, the system boundary differs slightly because manure-related emissions are not attributed to milk production processes in the cut-off version of the *ecoinvent* database, but rather to manure consumers as mentioned in Section 3.1.2. However, it is assumed that the historical learning observed in milk production systems can still be transferred to the milk production processes in *ecoinvent* for the estimation of prospective climate change impacts, as both represent the same underlying production technology.

3.5.2 Construction of decarbonization pathways under learning effects

In order to develop prospective climate change impacts based on historical learning effects, a two-step approach is adopted.

First, prospective impacts are estimated for the processes identified as subject to learning effects, namely milk and potato production. To link environmental learning with temporal dynamics, future production growth is used to derive the technology doubling time, a parameter representative of the rate of evolution of experience in the future. Projected production quantities are obtained from the latest OECD-FAO Agricultural Outlook (2025–2034) [7]. For potato production, the category “Roots and Tubers” is used as a proxy, as potatoes account for the largest share of global production within this category. For milk production, the category “Raw milk from bovine animals” is used.

To project environmental learning over time, the production-based learning relationship derived from Wright’s law is reformulated as a function of time. This transformation relies on the assumption that production follows an exponential growth trajectory characterized by a constant annual growth rate.

The projected production growth over the period 2025–2034 is converted into an annualized growth rate using the compound annual growth rate (CAGR):

$$\text{CAGR}_{l(p)} = \left(\frac{P_{l(p)}^{2034}}{P_{l(p)}^{2025}} \right)^{\frac{1}{9}} - 1 \quad (3.17)$$

where $P_{l(p)}^{2025}$ and $P_{l(p)}^{2034}$ are the projected annual production quantities of product $l(p)$ in 2025 and 2034, respectively. $l(p)$ refers to the product experiencing learning effects for product p (i.e. milk for cheese product and potato for potato product).

In order to link the learning effects with temporal dynamics, the technology doubling time is calculated :

$$T_{d,l(p)} = \frac{\ln(2)}{\ln(1 + \text{CAGR}_{l(p)})} \quad (3.18)$$

The doubling time is directly linked to the environmental learning rate, as it represents the time needed to double the level of accumulated experience, after which environmental impacts decrease by a factor determined by the learning rate. Starting from a baseline in 2025, the climate change impact of the learning processes $l(p)$ for product p is projected as follows:

$$CC_{l(p)}^p(t) = CC_{l(p),0}^p (1 - LR_{l(p)})^{\frac{t-t_0}{T_{d,l(p)}}} \quad t \geq t_0 \quad (3.19)$$

where $CC_{l(p),0}^p$ denotes the baseline climate change impact of process $l(p)$ at $t_0 = 2025$ for product p , and $LR_{l(p)}$ is the environmental learning rate estimated from historical data.

As discussed in Section 3.5.1, this formulation is applied only to the learning processes $l(p)$, i.e. the direct emissions associated with milk and potato production identified in the baseline results (i.e. climate change impacts from the original *ecoinvent* database).

In a second step, the projected impacts of the learning processes are combined with the evolution of the remaining system, which follows the SSP1-PkBudg650 background pathway. For a given product p , the climate change impact associated with all processes except the learning process is denoted by $CC_{\text{rest, SSP1-PkBudg650}}^p(t)$, while the climate change impact of the corresponding learning process is denoted by $CC_{l(p)}^p(t)$. The evolution of the total climate change impact of product p is therefore given by:

$$\begin{aligned} CC^p(t) &= CC_{l(p)}^p(t) + CC_{\text{rest, SSP1-PkBudg650}}^p(t) \\ &= CC_{l(p),0}^p (1 - LR_{l(p)})^{\frac{t-t_0}{T_{d,l(p)}}} + CC_{\text{rest, SSP1-PkBudg650}}^p(t) \end{aligned} \quad (3.20)$$

Alternatively, this expression can be rewritten in terms of the reduction induced by learning effects. The improvement due to learning is defined as:

$$\Delta CC_{l(p)}^p(t) = CC_{l(p),0}^p - CC_{l(p)}^p(t) = CC_{l(p),0}^p \left[1 - (1 - LR_{l(p)})^{\frac{t-t_0}{T_{d,l(p)}}} \right].$$

The total climate change impact of product p can then be expressed as the climate change impact under the SSP1-PkBudg650 scenario for the full system, minus the learning-induced reduction:

$$\begin{aligned} CC^p(t) &= CC_{\text{SSP1-PkBudg650}}^p(t) - \Delta CC_{l(p)}^p(t) \\ &= CC_{\text{SSP1-PkBudg650}}^p(t) - CC_{l(p),0}^p \left[1 - (1 - LR_{l(p)})^{\frac{t-t_0}{T_{d,l(p)}}} \right] \end{aligned} \quad (3.21)$$

3.6 Relating environmental impacts to planetary boundaries

In order to assess whether prospective improvements in climate change impacts are sufficient to remain within the planetary boundary for carbon, the allocated share of the safe operating space (soSOS) attributable to the food products considered in this study is computed.

The safe operating space (SOS) is defined as the difference between the planetary boundary and the natural background level, corresponding to pre-industrial environmental pressures. Based on this definition, this section describes how the climate change SOS is allocated to the functional units of the studied food products.

3.6.1 Downscaling planetary boundaries to a national carbon budget

Two main approaches exist to relate environmental impacts to planetary boundaries. (I) PB-LCIA, in which Life Cycle Impact Assessment (LCIA) indicators are adapted to ecological thresholds defined by planetary boundaries, and (II) LCIA-based planetary boundaries, in which planetary boundaries are translated into LCIA-compatible impact metrics. In this study, approach (II) is applied, enabling a direct comparison between climate change impacts and the allocated share of SOS, provided that the EF impact assessment method is used.

Based on Sala et al., the LCIA-consistent planetary boundary for climate change corresponds to 6.81×10^{12} kg CO₂e at the global level [18]. Since Denmark is used as the reference country, this global SOS is downscaled to a national carbon budget.

Several allocation principles can be used to distribute the global boundary at the national level, including equal per capita allocation (EpC), grandfathering based on historical emissions, and needs-based allocation accounting for differences in development and consumption needs. In this study, the EpC approach is selected, as it is the most widely used and transparent method. The Danish safe operating space for climate change is therefore computed as:

$$soSOS_{CC, DK} = SOS_{CC, World} \cdot \frac{P_{DK}}{P_{World}} \quad (3.22)$$

where $SOS_{CC, World}$ is the global safe operating space for climate change, P_{DK} and P_{World} denote the population of Denmark and the world, respectively, based on World Bank World Development Indicators [44].

3.6.2 Allocation of national carbon budget to product functional units

The national carbon budget is further allocated to the functional units of potato and cheese products using allocation principles. As there is no single correct way of allocating the SOS to a product, two allocation methods are applied: economic allocation (EC) and grandfathering allocation (GF). This allows for an assessment of the sensitivity of the results to the allocation choice, in line with the draft guidance on AESA from the JRC [45]. Both allocations are performed for the following functional unit: 1 kg of product at point of consumption in Denmark.

The GF approach allocates the budget based on the current share of emissions attributable to each product relative to total Danish annual consumption-based emissions:

$$soSOS_p^{GF} = soSOS_{DK} \cdot \frac{CC_p}{CC_{DK}} \quad (3.23)$$

where CC_p represents the climate change impact of 1 kg of product p calculated using *ecoinvent* with the EF 3.1 impact assessment method, and CC_{DK} denotes the total consumption-based climate footprint of Denmark in 2024, obtained from the European Consumption Footprint Platform [46].

The economic allocation distributes the budget according to the share of expenditure associated with each product relative to total annual household final consumption expenditure:

$$soSOS_p^{EC} = soSOS_{DK} \cdot \frac{\pi_p}{FCE_{DK}^{HH} \cdot N_{HH}} \quad (3.24)$$

where π_p is derived from consumer-reported retail prices in Denmark for product p from Numbeo [47], FCE_{DK}^{HH} corresponds to household final consumption expenditure in Denmark in 2024, based on the household budget survey [48], and N_{HH} is the total number of households in Denmark in 2024, based on the Civil Registration System [49].

3.6.3 Learning-based pathways towards absolute sustainability

The environmental learning theory introduced in Section 3.5.2 can be used to determine absolute sustainability pathways, i.e., pathways in which the climate change impact in 2050 is lower than the allocated SOS for a given product. Using equations describing the evolution of climate change impacts under learning effects (Equation 3.21), the learning rate required to achieve absolute sustainability for climate change in 2050 is calculated.

In this section, the learning rate is defined at the product level (i.e., cheese and potato production systems). This allows potential learning effects beyond the learning processes $l(p)$ considered in this study to be accounted for, for instance in processes such as cheese processing.

The climate change impact of product p in 2050, considering learning at the system level, is expressed as:

$$CC^p(2050) = CC_{SSP1-PkBudg650}^p(2050) - CC_0^p \left(1 - (1 - LR_p)^{\frac{25}{T_{d,p}}} \right) \quad (3.25)$$

with CC_0^p the baseline climate change impact of product p , LR_p the system-level learning rate for product p , and $T_{d,p}$ the technology doubling time for product p . For cheese, since learning is assumed to occur at the system level rather than being limited to milk production, the doubling time is recalculated based on the cheese production growth rate from the OECD-FAO Agricultural Outlook (2025–2034) for the product “Cheese, fresh or processed” [7].

The hypothetical learning rate required for product p to achieve absolute sustainability in 2050 is derived by imposing that the climate change impact in 2050 is lower than the least stringent safe operating space constraint ($soSOS^{GF}$ for cheese and $soSOS^{EC}$ for potato):

$$CC^p(2050) = soSOS_p,$$

which yields:

$$LR_{p,abs} = 1 - \left[1 - \frac{CC_{SSP1-PkBudg650}^p(2050) - soSOS^p}{CC_0^p} \right]^{\frac{T_{d,p}}{25}}. \quad (3.26)$$

where $CC_{SSP1-PkBudg650}^p(2050)$ is the climate change impact of product p under the background scenario in 2050, CC_0^p is the baseline climate change impact of product p in 2025, $soSOS^p$ is the allocated safe operating space for product p for a given allocation method, LR_p is the system-level learning rate, and $T_{d,p}$ is the technology doubling time.

In order to compare this learning rate with the learning rate implied by the scenario-based and learning-based approaches for prospective climate change impacts, an equivalent learning rate is computed for both approaches based on Equation 3.25, with $CC_m^p(2050)$ representing the prospective climate change impacts of method m (scenario-based or learning-based) in 2050. The equivalent learning rate is then expressed as:

$$LR_{p,m} = 1 - \left[1 - \frac{CC_{SSP1-PkBudg650}^p(2050) - CC_m^p(2050)}{CC_0^p} \right]^{\frac{T_{d,p}}{25}}. \quad (3.27)$$

Comparing the equivalent learning rate of the prospective decarbonization pathways modeled in this study, $LR_{p,m}$, with the learning rate required to reach absolute sustainability, $LR_{p,abs}$, allows the identification of the extent of additional learning required to close the absolute sustainability gap.

3.7 Sensitivity analysis

3.7.1 Influence of IAM scenario selection

Both prospective approaches used in this study rely on modeling background improvements using a single IAM scenario, REMIND SSP1-PkBudg650.

To assess the sensitivity of the results to the choice of background scenario, an alternative IAM scenario is selected and the evolution of climate change impacts between 2025 and 2050 for cheese and potato production is compared. As discussed in Section 3.3.1, IMAGE is another IAM relevant to this study. To ensure a comparable level of decarbonization to that achieved under SSP1-PkBudg650, the IMAGE scenario associated with the lowest GMST increase by 2100 is selected: SSP1-VLLO. This scenario follows the same SSP1 socio-economic pathway as the scenario used in this study but relies on a different set of mitigation measures, represented by the VLLO (Very Low Overshoot) climate target, which is projected to lead to a GMST increase of 1.38°C in 2100 [24].

3.7.2 Influence of learning boundary definition

The learning rate is a key parameter in the learning-based projection of climate change impacts from historical trends developed in Section 3.5.

The learning boundary is defined here as the set of emissions from the cheese and potato systems that are assumed to be affected by environmental learning effects on climate change impacts. In this study, the learning boundary is restricted to direct emissions from milk and potato production in order to remain consistent with the scope of the FAO dataset from which the empirical learning rate was estimated.

However, part of the observed learning effects is likely driven by yield improvements (e.g., kg of milk per cow and kg of potatoes per hectare), which may also reduce upstream input requirements as a consequence of higher productivity.

Therefore, a sensitivity analysis is conducted to assess the influence of the choice of the learning boundary on the prospective climate change results. More specifically, the first sensitivity scenario extends the learning boundary to first-tier inputs of milk and potato production, such as feed crops for dairy cows and tillage operations. In a second sensitivity

scenario, a more optimistic configuration is considered in which learning is applied to all processes within cheese and potato production systems, thereby including, for instance, potential learning effects in the cheese processing stage. In these sensitivity scenarios, only the learning boundary is modified, while the learning rate remains identical to that used in the baseline case.

The prospective climate change impacts are then recalculated based on the new learning boundaries using Equation 3.20, with an adjusted definition of $l(p)$ consistent with each learning boundary specification.

The growth rate used to compute the technology doubling time in these scenarios is adapted to the boundary definition to reflect the product driving learning effects. When the boundary includes milk and potato first-tier inputs, learning is assumed to be driven by experience in the milk and potato systems respectively. Therefore, the milk growth rate is used for milk-related processes and the potato growth rate for potato-related processes. When learning is applied to the full cheese system, experience is assumed to be driven by cheese production growth, and the cheese growth rate is used.

4 Results

4.1 Environmental impacts of potato and cheese systems under baseline conditions

4.1.1 Baseline life cycle impact assessment results

Figure 4.1 presents the environmental impacts of potato and cheese at the endpoint level using the single-score metric from EF 3.1 under baseline conditions. Baseline conditions correspond to the standard *ecoinvent* datasets, in which no prospective modifications are applied.

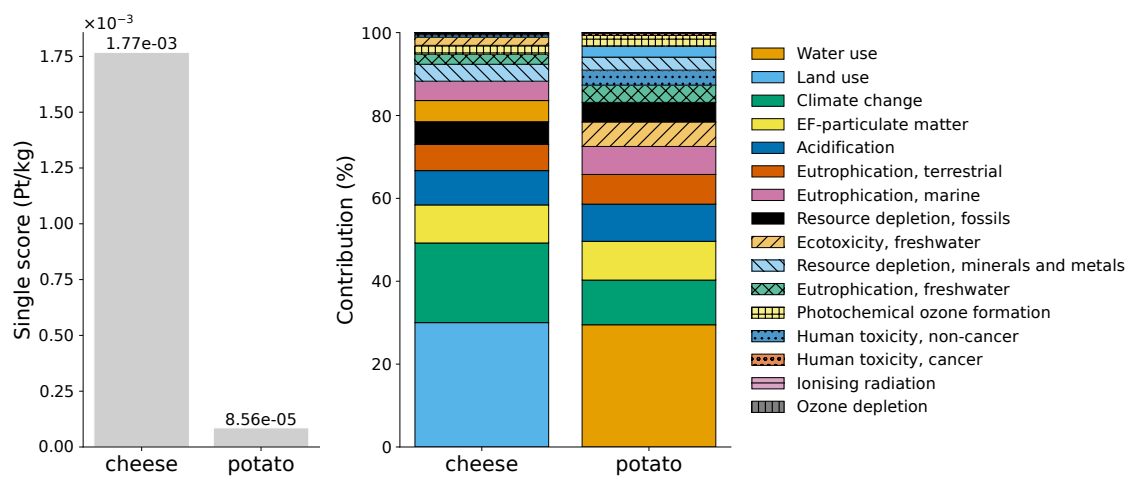


Figure 4.1: Baseline endpoint-level environmental impacts and contribution of each impact category to the single-score results. Impacts are calculated using *ecoinvent* 3.12.

Cheese exhibits a single-score environmental impact approximately 21 times higher than that of potatoes. This difference is primarily driven by the land use impact category, with a midpoint value of 5.47×10^3 (soil quality index, dimensionless). This impact is predominantly associated with cow milk production, which accounts for 91% of the total contribution within this category. The latter is mainly driven by land occupation and land transformation processes linked to livestock grazing systems.

For potatoes, the main environmental hotspot is the water use category, with a midpoint value of 3.41 m^3 world equivalent deprived. Irrigation during cultivation accounts for 97% of the total impact in this category.

Climate change is identified as the second most significant impact category for both products, with midpoint impacts of $12.2 \text{ kg CO}_2 \text{ eq}$ for cheese and $0.331 \text{ kg CO}_2 \text{ eq}$ for potatoes. At endpoint level, climate change contributes 19% of the total impact for cheese and 11% for potatoes, which supports the decision to focus on this impact category in the present study. Full midpoint results are provided in Appendix (Table 4).

4.1.2 Baseline climate change hotspot analysis

Figure 4.2 presents the ten largest contributors to climate change impacts for each product under baseline conditions. Additional information on the associated elementary flows and their roles within the cheese and potato production systems is provided in Table 4.1.

Cheese			
Process	Share	Description	Main elementary flows
Cow milk production	34.2%	Milk for cheese production and yoghurt production (with cheese as a by-product)	CH ₄ emissions from enteric fermentation
Electricity production	15.4%	Electricity for cheese production, milking machine and operation of the cow housing system	CO ₂ emissions from fuel combustion (coal, gas)
Cow feed production	13.7%	Crops for cow feeding (soybean, alfalfa-grass silage, maize grain)	N ₂ O emissions from fertilizer inputs
Land use change	7.1%	Clear-cutting of primary forest in South America for soybean production	CO ₂ emissions due to variation in carbon stock
Road transport	6.2%	Road transport to consumer by truck	CO ₂ emissions from diesel combustion
Potato			
Process	Share	Description	Main elementary flows
Potato production	17.2%	Potato cultivation	N ₂ O emissions from fertilizer inputs
Road transport	15.6%	Road transport to consumer by truck	CO ₂ emissions from diesel combustion
Electricity production	12.2%	Electricity for pesticides, agricultural machinery and seed production	CO ₂ emissions from fuel combustion (coal, gas)
Heat production	9.1%	Heat for ammonia and machinery production and steam generation for urea production	CO ₂ emissions from fuel combustion (coal, gas)
Ammonia production	6.8%	Ammonia used for fertilizer production (urea, ammonium nitrate)	CO ₂ emissions from coal combustion

Table 4.1: Top five contributors to climate change impacts under baseline conditions. The associated main elementary flows and the role of each process are described.

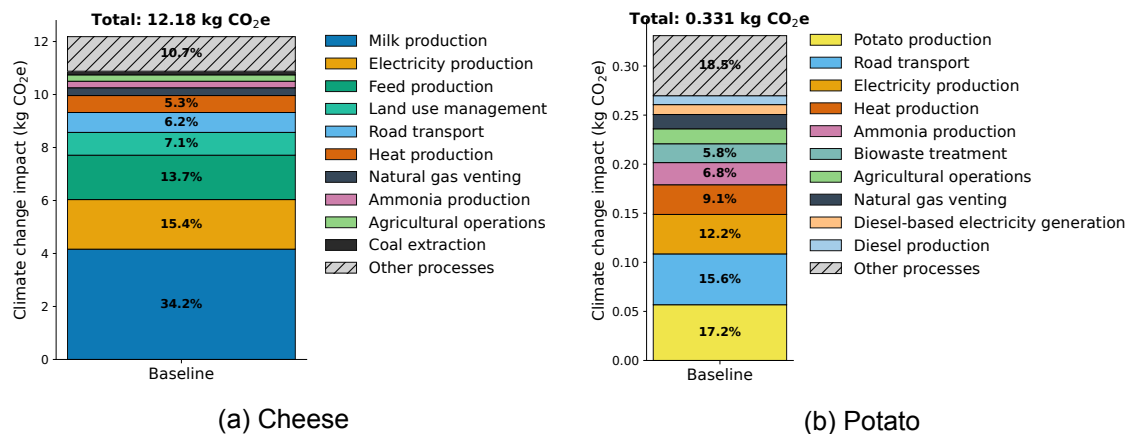


Figure 4.2: Top ten contributors to climate change impacts under baseline conditions, including their share of the total impact.

Most of the carbon footprint of cheese production originates from milk production and upstream impacts associated with feed crop cultivation, together accounting for 55% of total emissions. For potato production, the carbon footprint is more evenly distributed across hotspots, with background processes such as electricity generation, heat production, and transportation contributing 37% of total emissions, in addition to on-field emissions from potato cultivation.

The dominant elementary flows differ between the two products: CH_4 is the main contributor to cheese production impacts, accounting for 81% of direct emissions from milk production, whereas N_2O dominates potato production impacts, representing 96% of direct emissions from potato cultivation.

4.2 Prospective climate change impacts in 2050 under background scenarios

4.2.1 Climate change impact reductions from background system transformations

Figures 4.3 and 4.4 provide an overview of the climate change impact reductions achieved in 2050 under the background scenario SSP1-PkBudg650 relative to the baseline conditions. Processes are grouped according to the approach detailed in section 3.2.1. The contribution of each activity group to the overall reduction in climate change impacts is calculated as the difference between its impact in 2050 and in the baseline conditions, divided by the total baseline climate change impact, making the contribution summable across activity groups.

Overall, background transformations under the SSP1-PkBudg650 scenario reduce the carbon footprint of cheese production by 34.7%, from 12.18 to 7.96 kg CO_2e per kg product, and that of potato production by 47.8%, from 0.331 to 0.173 kg CO_2e .

Additional details on the mitigation measures implemented in the SSP1-PkBudg650 scenario and their contribution to the observed reductions are provided below:

- Electricity production: reduction in emissions related to electricity production, primarily driven by coal-fired power plants in China. This is achieved through an increase in plant efficiency (the efficiency of hard coal power plants in China reaches 37% in 2050) and a significant shift in the electricity mix towards renewable energy

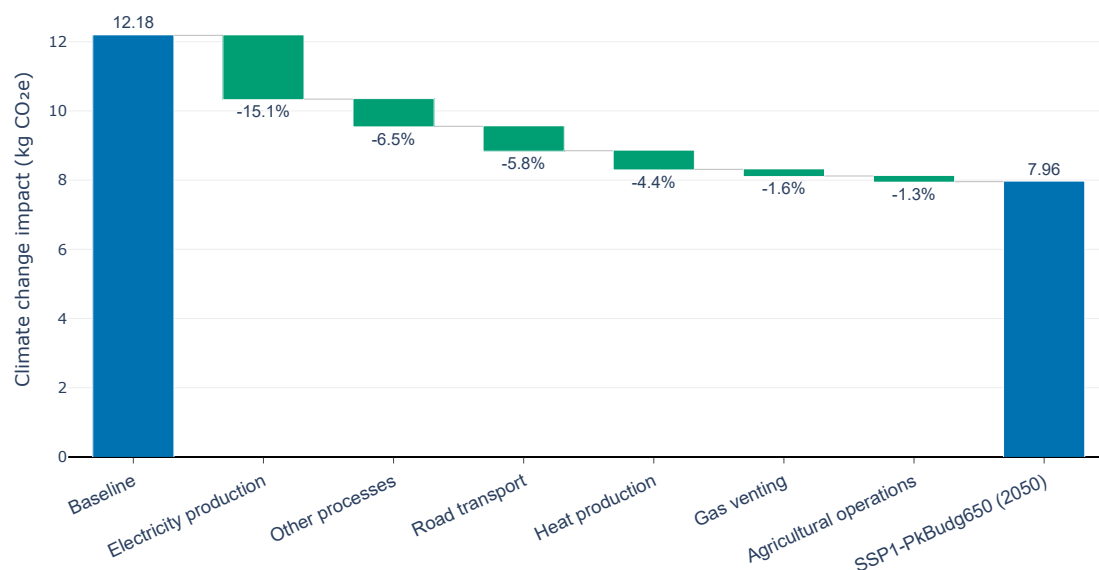


Figure 4.3: Contribution of background system transformations under SSP1-PkBudg650 in 2050 to climate change impact reductions for cheese. Only activity groups contributing more than 1% relative change compared to the baseline impact are shown.

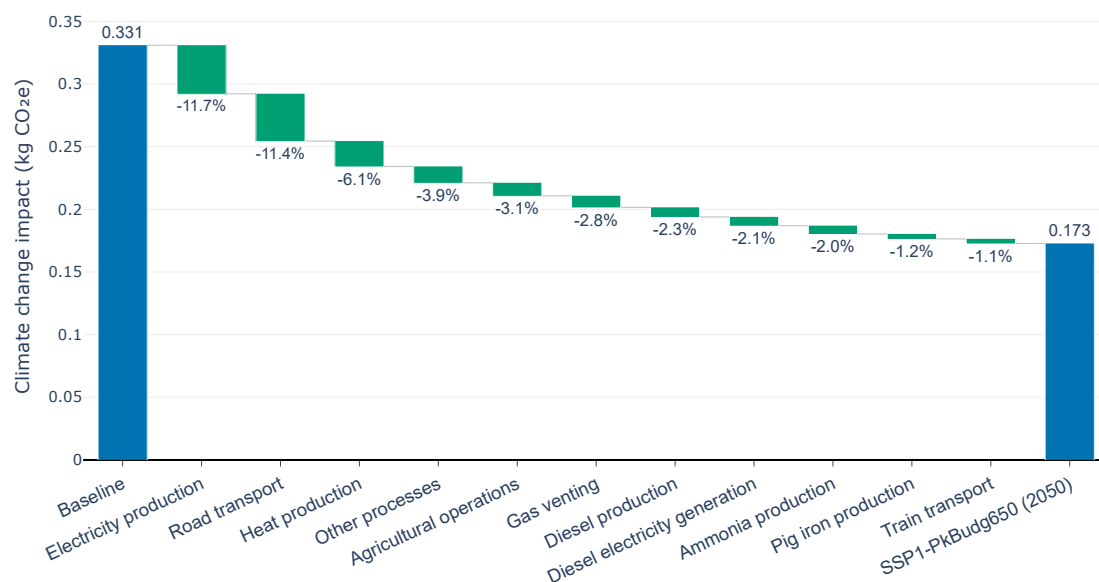


Figure 4.4: Contribution of background system transformations under SSP1-PkBudg650 in 2050 to climate change impact reductions for potato. Only activity groups contributing more than 1% relative change compared to the baseline impact are shown.

sources. Under this scenario, the global electricity mix in 2050 is projected to consist of 56% solar, 33% wind, and 7% hydro, while fossil-fuel-based generation accounts for less than 2%.

- Heat production: reduction in emissions related to heat production, mainly from hard coal-, fuel oil-, and natural gas-based heat generation. This is achieved through a transition towards water-to-water heat pumps and geothermal energy, each accounting for 50% of the heat production mix for district and industrial applications in 2050.
- Road transport: reduction in emissions related to road transport, primarily from diesel trucks. This reduction is driven by both a decrease in the carbon intensity of diesel fuels and a structural shift in the truck fleet towards low-emission technologies. The carbon intensity of diesel decreases as conventional fossil routes are progressively replaced by low-carbon synthetic pathways, including electrolysis-based hydrogen. This results in a 68% reduction in fossil CO₂ emissions from diesel combustion in transport. In parallel, the truck fleet shifts towards 95% battery electric vehicles, 4% diesel vehicles, and 1% fuel-cell and compressed gas vehicles.
- Gas venting: reduction in methane and carbon dioxide emissions during petroleum and natural gas extraction, driven by a strong decline in fossil fuel demand. Natural gas production is projected to decrease by 90% between 2025 and 2050 in this scenario.
- Agricultural operations: reduction in emissions related to diesel combustion in agricultural machinery. This is achieved through a decrease in the carbon intensity of fuels, which are increasingly sourced from lower-carbon production pathways, resulting in a 70% reduction in fossil CO₂ emissions from diesel combustion in agricultural machinery.
- Diesel production: reduction in emissions associated with diesel production due to the introduction of synthetic fuels and a decline in demand, as well as a phase-out of fossil-based diesel from petroleum refining. Non-synthetic diesel demand decreases by 80% between 2025 and 2050.
- Diesel electricity generation: reduction in emissions related to diesel combustion for electricity generation in applications such as refrigeration units. This is achieved through a reduction in fuel carbon intensity, similar to road transport, leading to a 70% decrease in fossil CO₂ emissions.
- Ammonia production: reduction in emissions related to ammonia production via steam reforming. This is driven by a decrease in the carbon intensity of natural gas inputs, with approximately 50% of the natural gas supply in 2050 coming from biomethane. Ammonia production via partial oxidation remains unchanged, as it relies on hard coal, whose emission intensity is not affected under this scenario.
- Pig iron production: reduction in emissions related to pig iron production, driven both by a reduction in the carbon intensity of fuels (notably biomethane) and by a shift in the steel production technology mix. New hydrogen-based production technologies progressively replace conventional coal-based steel production, accounting for more than 98% of the primary steel supply in 2050.
- Train transport: reduction in emissions from rail transport due to a decrease in diesel carbon intensity (69% reduction for diesel trains) and a modal shift towards electrification, with 96% of trains being electric. The remaining diesel-electric trains also

exhibit a 19% improvement in efficiency compared to 2020.

4.2.2 Residual climate change hotspots after background decarbonization

As a result of these background transformations, the climate change hotspot profile changes substantially. Figure 4.5 presents the top 10 contributors to climate change impacts in 2050 under background system improvements.

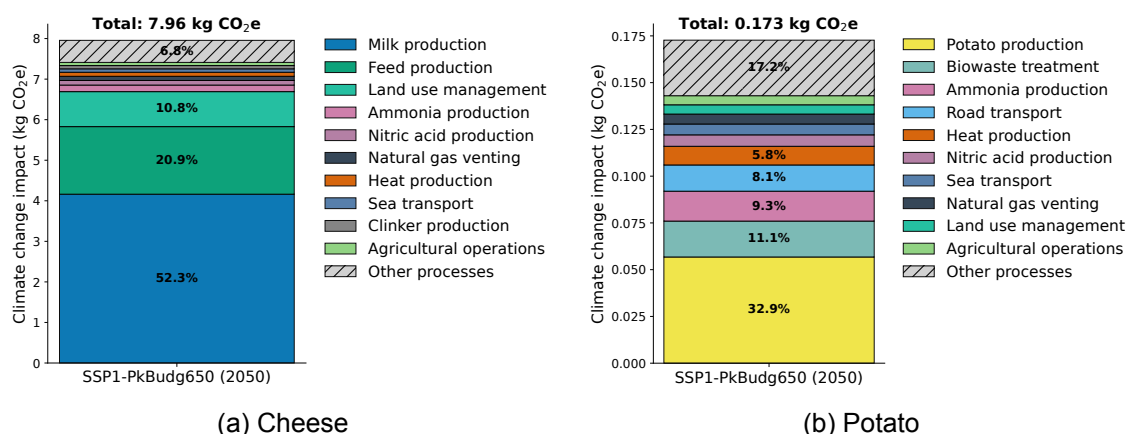


Figure 4.5: Top ten contributors to climate change impacts in 2050 under the SSP1-PkBudg650 scenario, including their share of the total impact.

Impacts associated with background activities such as electricity generation, heat production and transportation, decrease by 92% for cheese and 75% for potato. Consequently, previously secondary processes such as nitric acid production and biowaste treatment emerge as significant hotspots.

For cheese production, direct emissions from milk production, feed crop cultivation, and associated land-use change remain unaffected by the background scenario and together account for more than 84% of total climate change impacts in 2050. Similarly, direct emissions from potato cultivation remain unchanged and represent more than 32% of total impacts. These findings justify focusing the foreground scenarios on these key agricultural hotspots in order to complement background decarbonization efforts and achieve further reductions in prospective climate change impacts.

Additional information on the main contributing processes that exhibit less than 1% improvement relative to the baseline scenario is provided in Table 4.2, together with potential mitigation measures that could complement background decarbonization and further reduce their climate change impacts.

4.2.3 Environmental trade-offs under background scenarios

This section examines the variation in environmental impacts between the baseline and 2050 scenarios in order to identify potential burden-shifting effects introduced by the SSP1-PkBudg650 scenario. Figures 4.6 and 4.7 present the relative variation in each EF 3.1 impact category, together with the variation in the endpoint single-score results.

Under the SSP1-PkBudg650 scenario, impacts decrease in 11 out of 16 impact categories for cheese production and in 9 out of 16 impact categories for potato production by 2050. The largest reductions are observed for *resource use*, *fossils*, driven by the lower consumption of fossil resources such as petroleum, natural gas, and hard coal; *ionising radiation*, due to the reduced contribution of nuclear power in the future electricity mix;

Cheese			
Process	Share	Description	Mitigation strategies
Cow milk production	52.3%	Milk for cheese production and yoghurt production (with cheese as a co-product)	Implement dietary changes, supplement CH ₄ inhibitors or electron sinks
Cow feed production	20.9%	Crops for cow feeding (soybean, alfalfa-grass silage, maize grain)	Increase nitrogen use efficiency
Land use change	10.8%	Clear-cutting of primary forest in South America for soybean production	Reduce soybean share in feed, modify procurement strategies
Nitric acid production	1.4%	Intermediate product for ammonium nitrate fertilizer production and cleaning agent for cheese production	Reduce fertilizer use, implement N ₂ O abatement catalysts

Potato			
Process	Share	Description	Mitigation strategies
Potato production	32.9%	Potato cultivation	Increase nitrogen use efficiency
Biowaste treatment	11.1%	Treatment of potato waste generated during transportation	Switch to alternative treatment methods to prevent CH ₄ emissions from open dumping
Nitric acid production	3.5%	Intermediate product for ammonium nitrate fertilizer production	Reduce fertilizer use, implement N ₂ O abatement catalysts
Potato seed production	3.0%	Production of seed potatoes	Increase nitrogen use efficiency

Table 4.2: Main climate change hotspots exhibiting less than 1% variation in impact between the baseline and SSP1-PkBudg650 (2050), along with potential additional mitigation measures.

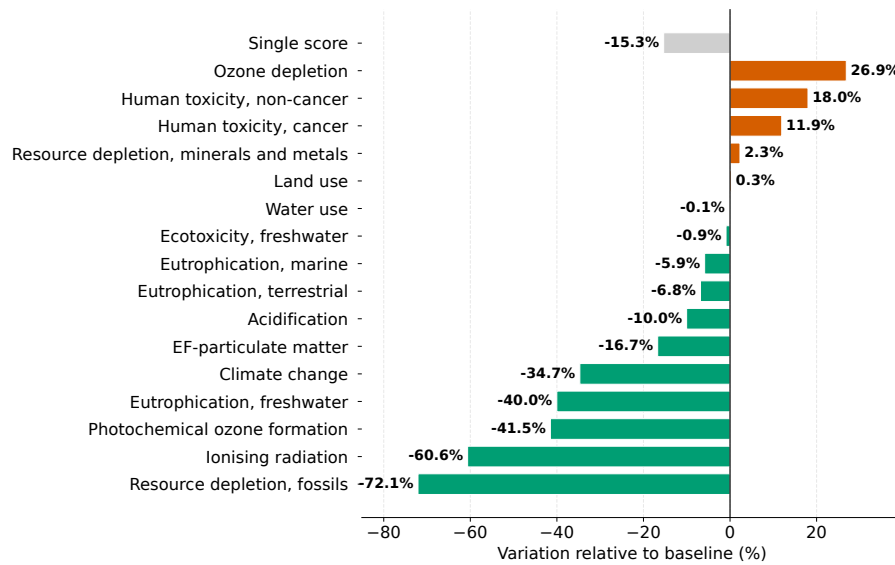


Figure 4.6: Variation in environmental impacts relative to baseline conditions under SSP1-PkBudg650 (2050) across impact categories for cheese. The corresponding single-score variation is also shown.

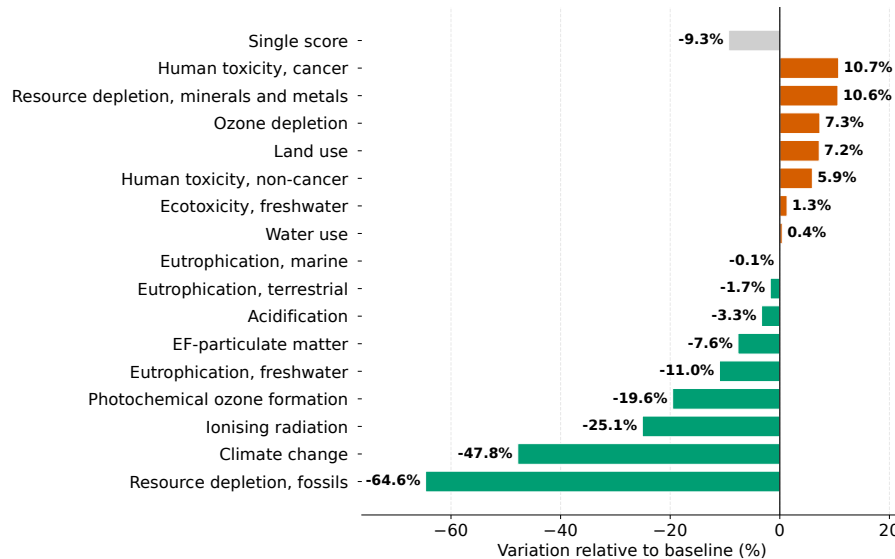


Figure 4.7: Variation in environmental impacts relative to baseline conditions under SSP1-PkBudg650 (2050) across impact categories for potato. The corresponding single-score variation is also shown.

photochemical ozone formation, resulting from lower gas venting and reduced NO_x emissions from transportation; and *climate change*, for which the main drivers were discussed in Section 4.2.1.

Despite these improvements, significant burden-shifting effects are observed for both products, particularly in the *ozone depletion* and *human toxicity* impact categories. The processes exhibiting the largest increases in impact between the baseline and 2050 scenarios, and thus contributing most to these burden-shifting effects, are reported in the Appendix (Table 5).

For cheese production, the most pronounced burden shifting occurs in the ozone depletion category, primarily driven by the increased production of methyl chloride, which is used in the manufacturing of electrolytes for lithium-sulfur battery systems deployed in electric trucks. Human toxicity impacts increase for both products in both the carcinogenic and non-carcinogenic categories. These increases are mainly associated with the additional production of metals such as copper, required for the expansion of electricity transmission and distribution infrastructure due to the massive electrification occurring in the SSP1-PkBudg650 scenario. In addition, the land use category for potato production exhibits a burden shift of 7.2%, largely attributable to increased logging activities associated with hydrogen production through woody biomass gasification.

Nevertheless, despite the increases observed in several midpoint categories, the end-point single-score results decrease for both products, indicating an overall environmental improvement under the SSP1-PkBudg650 scenario (-15.3% for cheese production and -9.3% for potato production).

4.3 Climate change mitigation potential of agri-food foreground scenarios

4.3.1 Climate change mitigation potential of individual agri-food scenarios

Agri-food scenarios are combined with the SSP1-PkBudg650 background scenario to derive prospective climate change impacts that include structural changes in the agricultural sector, in addition to technological progress in the rest of the economy. Table 4.3 reports the 2050 climate change impacts associated with each agri-food scenario combined with the SSP1-PkBudg650 background scenario, together with the percentage variation relative to the 2050 impacts of the SSP1-PkBudg650 scenario alone.

Compared with the climate change impacts obtained under background improvements only, the additional reductions range from -13.65% (NOP) to +3.81% (FL) for cheese production and from -19.15% (NI) to -4.37% (ET) for potato production.

Two mitigation scenarios lead to higher climate change impacts than under the SSP1-PkBudg650 scenario alone for cheese production: FR and FL. In the FR scenario, reductions in nitrogen fertilizer application and associated N₂O emissions are offset by lower crop yields. Consequently, climate change impacts related to feed crop cultivation and land management increase by 10% relative to the 2050 SSP1-PkBudg650 scenario, leading to an overall increase of 2.73% in total cheese climate change impacts. Similarly, in the FL scenario, reductions in enteric CH₄ emissions and improvements in milk productivity are outweighed by the additional upstream burdens associated with increased feed production, leading to a net increase of 3.81% in total climate change impacts.

Overall, machinery electrification scenarios provide the smallest additional reductions. This is because agricultural operations contribute only a limited share of total climate

Scenario group	Label	Impact (kg CO ₂ e)	Variation
Cheese			
Increased feeding level	FL	8.26	+3.81%
3-NOP supplementation	NOP	6.87	-13.65%
Nitrate supplementation	NS	7.21	-9.38%
Optimized fertilizer rate	FR	8.17	+2.73%
Optimized fertilizer placement	FP	7.30	-8.28%
Use of nitrification inhibitors	NI	7.14	-10.29%
Use of urease inhibitors	UI	7.51	-5.63%
Use of double inhibitors	NI+UI	7.48	-6.03%
Tractor electrification	ET	7.83	-1.54%
Potato			
Optimized fertilizer rate	FR	0.153	-11.51%
Optimized fertilizer placement	FP	0.152	-11.91%
Use of nitrification inhibitors	NI	0.140	-19.15%
Use of urease inhibitors	UI	0.155	-10.43%
Use of double inhibitors	NI+UI	0.150	-13.13%
Tractor electrification	ET	0.165	-4.37%

Table 4.3: Prospective climate change impacts in 2050 by scenario and variation relative to the SSP1-PkBudg650 (2050) background scenario. Bold values indicate the best-performing scenario within each scenario group. Colours denote scenario groups: enteric fermentation mitigation, agricultural direct emissions mitigation, and agricultural machinery emissions mitigation.

change impacts in the system under baseline conditions, accounting for 2.0% of impacts for cheese production and 4.6% for potato production. Furthermore, the impacts associated with these operations have already been substantially reduced under the SSP1-PkBudg650 background scenario, with a decrease of approximately 67% for both products by 2050 due to a reduction in diesel carbon intensity.

Several mitigation measures also require the production of additional inputs. This is the case for the NS and NOP scenarios, where nitrate supplements and 3-NOP introduce additional upstream burdens compared with the baseline situation. However, these contributions remain small relative to total climate change impacts. Under the NOP scenario, the production of 3-NOP accounts for only 0.25% of total climate change impacts in 2050, while nitrate production contributes 0.61% under the NS scenario.

The most promising mitigation options were subsequently combined to explore the lowest climate change impacts that could realistically be achieved under the considered scenarios, while still being combined with the SSP1-PkBudg650 background scenario. The best-performing scenario from each group was therefore selected (NI, ET, and NOP for cheese production; NI and ET for potato production) and combined.

Figures 4.8 and 4.9 present the cumulative reductions in climate change impacts relative to the baseline situation for these most promising mitigation options.

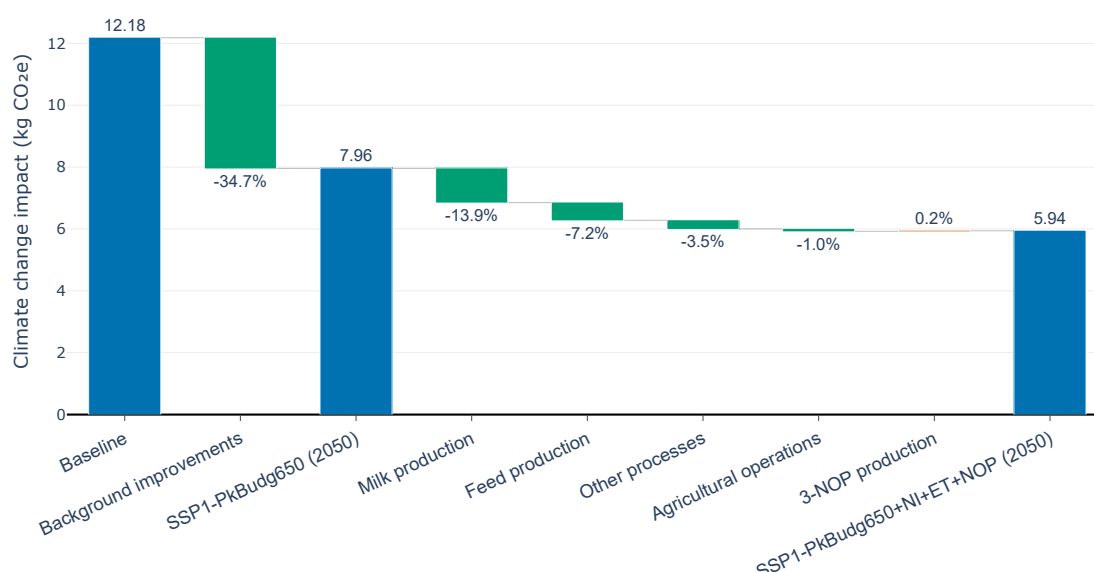


Figure 4.8: Contribution of background system transformations (SSP1-PkBudg650) combined with agri-food scenarios NI, ET, and NOP in 2050 to climate change impact reductions for cheese.

The substantial reductions already achieved through the SSP1-PkBudg650 background scenario are visible, corresponding to decreases of 34.7% for cheese production and 47.8% for potato production.

When the selected mitigation measures are combined, climate change impacts decrease to 5.94 kg CO₂e for cheese production and 0.133 kg CO₂e for potato production. This corresponds to an additional reduction of 25.3% relative to SSP1-PkBudg650 for cheese and 23.0% for potato, resulting in total reductions of 51.2% and 59.8%, respectively, compared with the baseline.

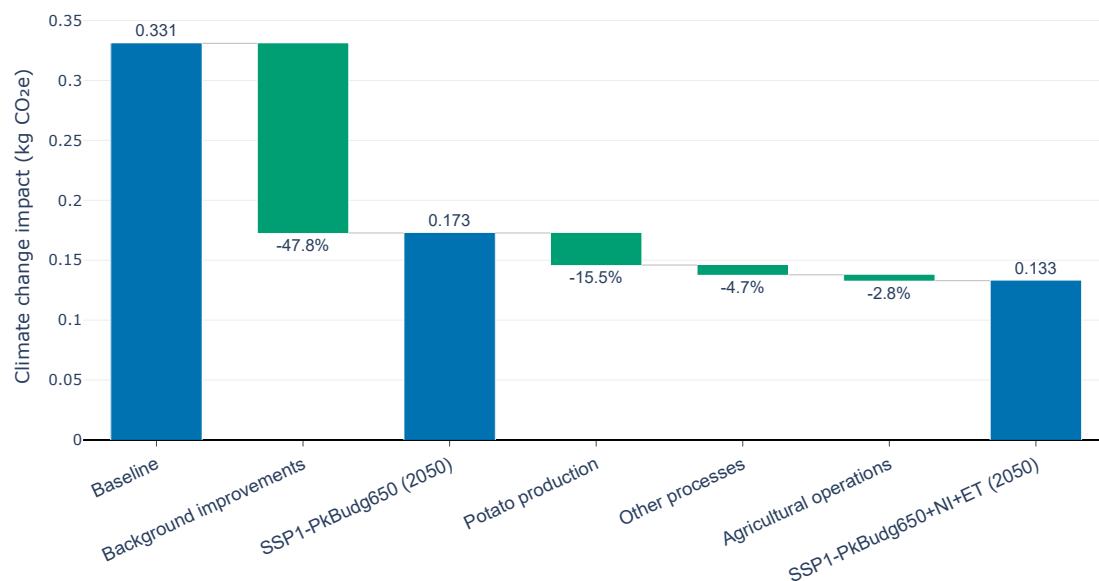


Figure 4.9: Contribution of background system transformations (SSP1-PkBudg650) combined with agri-food scenarios NI and ET in 2050 to climate change impact reductions for potato.

For cheese production, the largest additional reduction is achieved through 3-NOP supplementation, which lowers enteric CH₄ emission intensity and reduces climate change impacts by a further 13.9% relative to SSP1-PkBudg650. This is followed by reductions associated with nitrification inhibitors, which decrease emissions from feed crop cultivation by 7.2%, and by the adoption of electric tractors, which lowers emissions from agricultural operations by 1.0%. Additional gains arise from reduced input requirements and biosphere emissions associated with higher feed crop yields and improved milk productivity, contributing a further 3.4% reduction.

For potato production, the largest improvement is obtained through nitrification inhibitors, which reduce emissions from crop cultivation and lower climate change impacts by an additional 15.5% relative to SSP1-PkBudg650. The adoption of electric tractors contributes a further 2.8% reduction through lower emissions from agricultural operations. Additional improvements are achieved through higher potato yields, which reduce input requirements and associated biosphere flows, resulting in a further 4.7% reduction.

4.3.2 Residual climate change hotspots under combined mitigation scenarios

Figure 4.10 presents the top 10 contributors to the climate change impacts in 2050 after combining the most effective mitigation measures identified for each product.

For cheese production, direct emissions associated with milk production, feed crop cultivation, and related land management activities still account for 81.2% of total climate change impacts. This highlights the limited mitigation potential of technological measures alone for dairy systems, as a large share of residual emissions originates from biological processes that are difficult to eliminate. Further reductions would require additional interventions targeting upstream agricultural inputs, particularly nitrogen fertilizer production. Ammonia and nitric acid production remain important contributors to the remaining footprint, suggesting that further decarbonization of fertilizer supply chains could provide

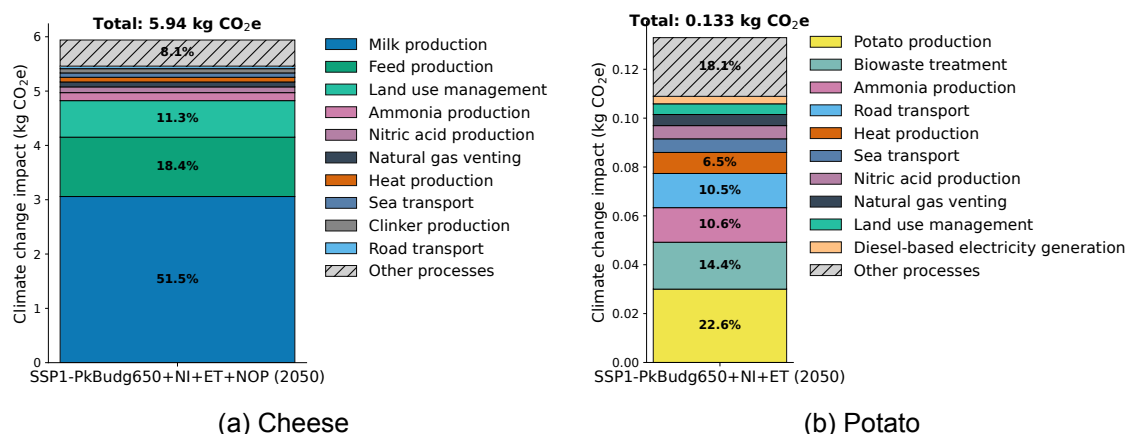


Figure 4.10: Top ten contributors to climate change impacts in 2050 under the combination of the SSP1-PkBudg650 scenario with the best-performing agri-food scenario, including their share of the total impact.

additional benefits.

For potato production, mitigation measures targeting crop cultivation substantially reduce its contribution to the total carbon footprint, with direct emissions from potato production accounting for only 22.6% of impacts in 2050. As a result, the remaining emissions are increasingly dominated by background processes that are only partially mitigated under the SSP1-PkBudg650 scenario. Road transport activities represent 10.5% of total climate change impacts, while ammonia production contributes a further 10.6%. Additional mitigation efforts should therefore focus on the decarbonization of these upstream processes, as well as on biowaste treatment activities, which remain a non-negligible source of greenhouse gas emissions.

4.4 Prospective climate change impacts under learning-based projections

4.4.1 Derivation of historical environmental learning rates

Figure 4.11 presents the fit of the Wright learning curve to the emission intensity profile in log-log space. The goodness of fit is assessed using the coefficient of determination (R^2), consistent with the linear formulation of the Wright curve in log-log space. The corresponding results are reported in Table 4.4.

Product	LR (%)	C_0	R^2
Milk	27.6	1.51	0.974
Cereals	19.6	0.173	0.731

Table 4.4: Wright curve parameters for milk and cereals learning

Learning effects are estimated using direct emissions from milk production for cheese, and direct emissions from cereals as a proxy for potato production (see Section 3.5.1 for more details). The Wright curve provides a strong fit for milk emission intensity as a function of cumulative production, with an R^2 of 0.974 in log-log space. The corresponding environmental learning rate is 27.6% for climate change impacts.

For cereals excluding rice, the fit is weaker but still shows a clear decreasing trend in

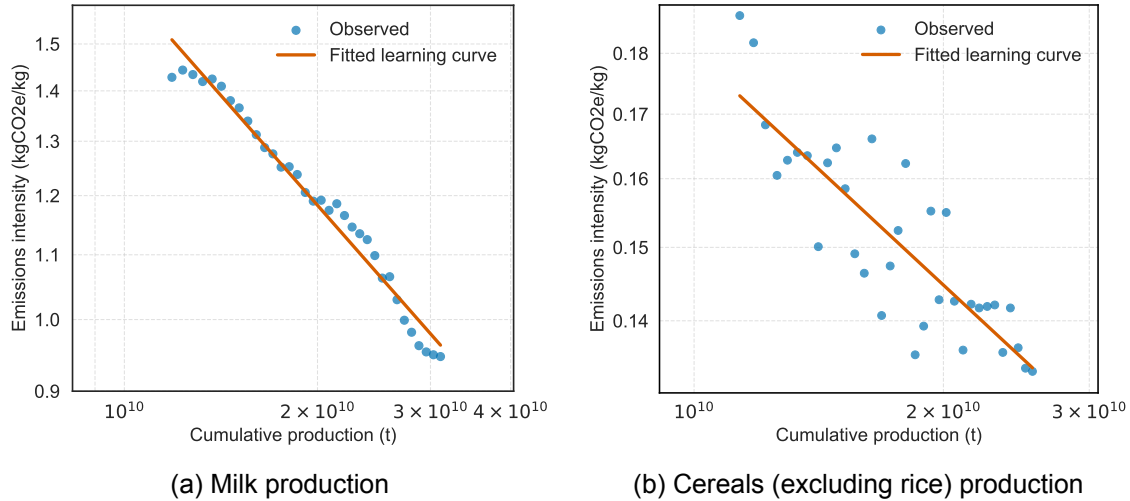


Figure 4.11: Wright learning curve fits for emission intensity as a function of cumulative production in log-log space. Cumulative production is computed starting in 1961, while the fit is performed using data from 1990 onwards.

emission intensity with cumulative production. The higher variability in the data results in an R^2 of 0.731 and an environmental learning rate of 19.6%.

4.4.2 Integrated projections combining learning effects and background transformations

Table 4.5 present the projected annual CAGR and the corresponding doubling time T_d for the different products :

Product	Parameter	Value	Unit
Milk	CAGR	1.80	%
	T_d	38.92	year
Cheese	CAGR	1.04	%
	T_d	66.78	year
Potato	CAGR	1.87	%
	T_d	37.40	year

Table 4.5: Compound annual growth rates and doubling times for selected food products.

Figures 4.12 and 4.13 present the integrated prospective climate change impacts combining learning effects and background changes under the SSP1-PkBudg650 scenario.

The results are decomposed into three components: (i) direct emissions from milk and potato production, (ii) direct emissions from first-tier inputs for milk and potato production (e.g. feed crop production for milk, agricultural operations such as tillage and fertilizer-related emissions for potato), and (iii) all remaining processes. Learning effects are considered only for the component (i), as the determination of the learning rate was done on this boundary. The sensitivity of the results to the boundary of application of the learning rate is further described in Section 4.6.2.

Under learning effects, milk direct emissions decrease by 18.7% between 2025 and 2050. Emissions related to first-tier inputs decrease by 13.3% and other emissions decrease

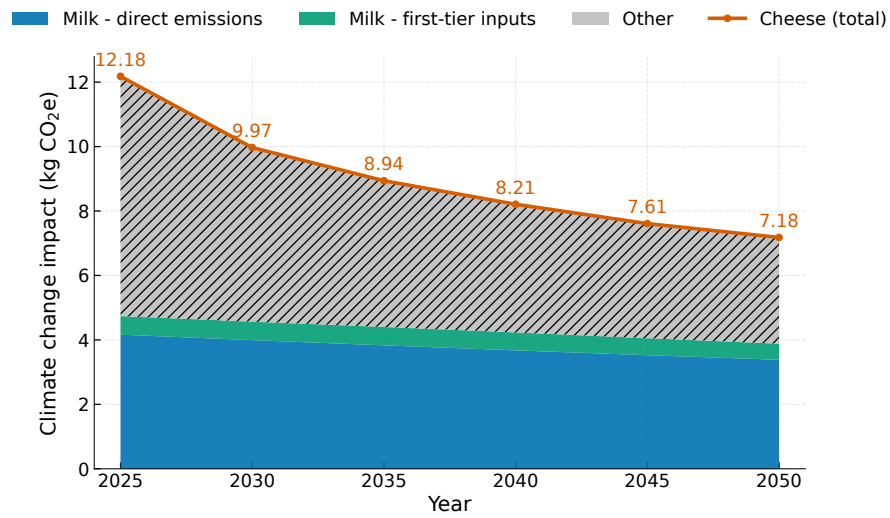


Figure 4.12: Learning-based prospective climate change impacts between 2025 and 2050 for cheese. Learning effects are applied to direct emissions from milk production, while the rest of the system evolves under the SSP1-PkBudg650 scenario.

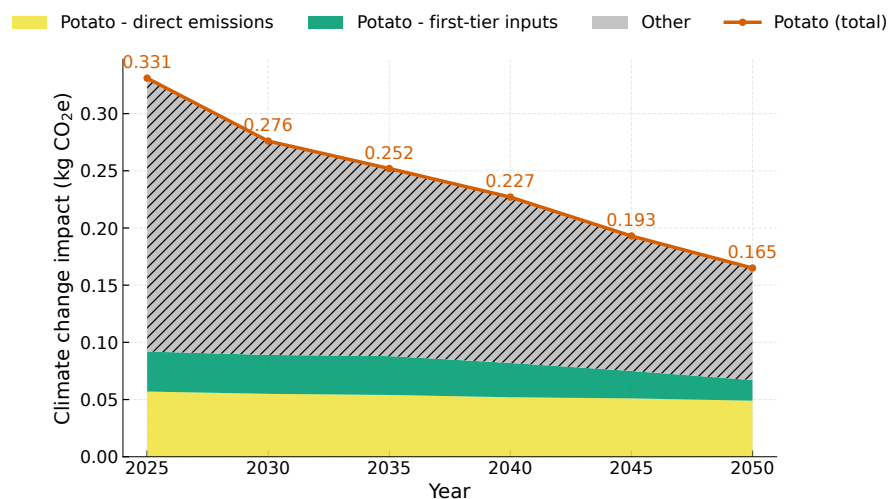


Figure 4.13: Learning-based prospective climate change impacts between 2025 and 2050 for potato. Learning effects are applied to direct emissions from potato production, while the rest of the system evolves under the SSP1-PkBudg650 scenario.

by 55.7% under the SSP1-PkBudg650 background scenario. Overall, the total climate change impact for cheese reaches 7.18 kg CO₂e, corresponding to a 41.1% reduction relative to the baseline.

For potato, direct emissions decrease by 13.6% due to learning effects, while emissions from first tier inputs decrease by 49% and other emissions decrease by 59.0% under background improvements. Overall, the total climate change impact reaches 0.165 kg CO₂e, corresponding to a 50.1% reduction relative to the baseline.

However, these reductions represent only a 9.8% decrease for cheese (4.5% for potatoes) compared to the improvements under SSP1-PkBudg650 alone, whereas the mitigation measures modeled in the scenario-based approach achieve an additional 25.3% reduction for cheese (23.0% for potatoes) relative to SSP1-PkBudg650 alone. This is due to the modest projected growth rates of the milk and potato sectors, which limit the development of learning effects.

4.5 Decarbonization pathways in relation to planetary boundaries

4.5.1 Allocation of the climate change safe operating space

Table 4.6 reports the SOS allocations for cheese and potato production systems, i.e., the share of the climate change planetary boundary attributed to each functional unit under different allocation principles.

Allocation method	Value (kgCO ₂ e)
Denmark - Annual budget	
Equal per capita	5.0×10^9
Cheese - budget for 1 kg (Denmark)	
Economic	0.46
Grand-fathering	1.15
Potato - budget for 1 kg (Denmark)	
Economic	0.070
Grand-fathering	0.031

Table 4.6: Safe operating space allocation results for climate change impacts in Denmark, including national budget and product-level allocations for cheese and potato systems.

For cheese, the allocated share of SOS is higher under the grand-fathering approach than under the economic allocation method, whereas the opposite trend is observed for potato. This difference is driven by the lower ratio of price to emissions intensity for cheese compared with potato (approximately 8 DKK/kg CO₂e for cheese versus 45 DKK/kg CO₂e for potato).

Figure 4.14 presents the exceedance ratio for each product and allocation method, defined as the ratio between the climate change impact and the allocated SOS. A system is considered to operate within its absolute sustainability limit when the exceedance ratio is below 1, indicating that impacts remain within the allocated safe operating space.

For both products and allocation methods, multiple scenarios are compared: (i) baseline impacts based on standard *ecoinvent* processes with no prospective modifications,

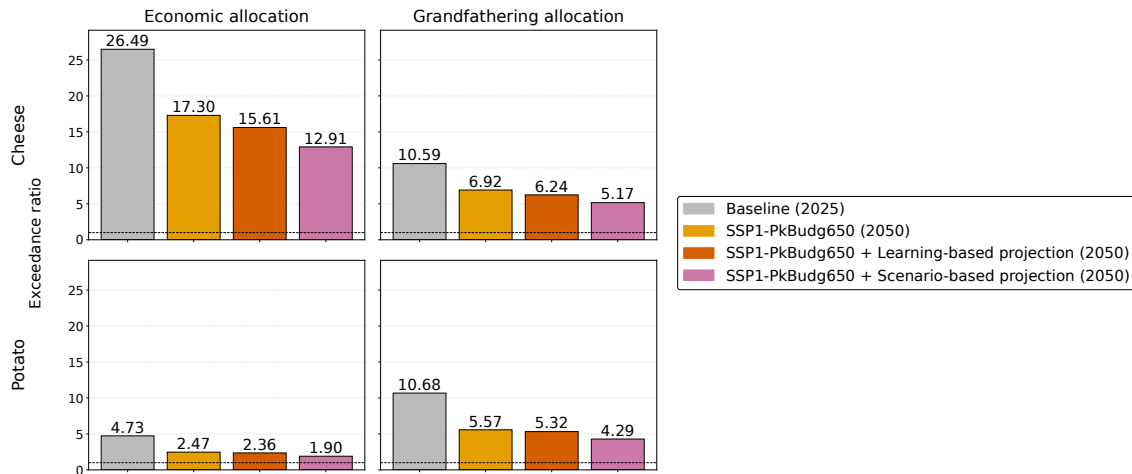


Figure 4.14: Exceedance ratios of climate change impacts relative to the SOS under different scenarios and allocation methods.

(ii) 2050 impacts under background improvements following SSP1-PkBudg650, (iii) 2050 impacts combining background improvements with foreground mitigation measures, and (iv) 2050 impacts combining background improvements with learning-based projections.

Neither product operates within its allocated safe operating space under any allocation method. Baseline exceedance factors range from 4.73 (potato, economic allocation) to 26.49 (cheese, economic allocation), highlighting a substantial sustainability gap under current production conditions.

Even after implementing mitigation measures across both background and foreground systems, or when accounting for impact reductions due to learning effects, the products remain outside their allocated SOS. For potato, the combination of the SSP1-PkBudg650 scenario with the foreground scenarios NI and ET in 2050 results in an exceedance factor of 1.90 (corresponding to an exceedance of +0.063 kg CO₂e above the absolute sustainability threshold). For cheese, the combination of the SSP1-PkBudg650 scenario with the foreground scenarios NI, ET, and NOP in 2050 results in an exceedance factor of 5.17 (+4.79 kg CO₂e).

4.5.2 Comparison of prospective decarbonization pathways

Scenario-based and learning-based trajectories are first compared within the restricted emission scope to which learning effects are applied, namely direct emissions from milk and potato production. This comparison provides insight into whether the mitigation measures implemented in the agri-food scenarios are more or less ambitious than the rates of improvement observed historically. Figures 4.15a and 4.15b illustrate the evolution of direct emissions from milk and potato production between 2025 and 2050 under both approaches.

For both products, the scenario-based trajectory remains below the learning-based trajectory throughout 2025-2050. This indicates that use of nitrification inhibitors for potato production and 3-NOP supplementation for milk production outperform historical improvement trends, meaning that under the assumptions of production growth rate considered in this study, the reduction potential of the measures considered in the scenario-based approach can be considered ambitious compared to what have been achieved in the past.

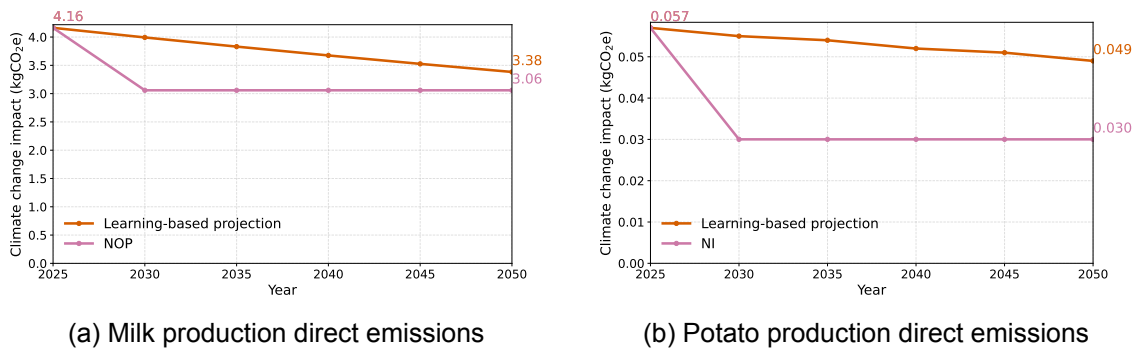


Figure 4.15: Evolution of climate change impacts from direct emissions of milk and potato production between 2025 and 2050 under the learning-based and scenario-based approaches. For the scenario-based approach, the best-performing scenario affecting this emission scope is considered (NOP for milk and NI for potato).

Then, three prospective decarbonization pathways between 2025 and 2050 for the full potato and cheese production systems are compared (Figures 4.16 and 4.17). Two pathways correspond to the methodological approaches explored in this study: (i) a scenario-based pathway combining background improvements with the most effective foreground mitigation measures (NI+ET+NOP for cheese and NI+ET for potato, as detailed in Section 3.4), and (ii) a learning-based pathway derived from a projection of historical learning effects (Section 3.5), combined with SSP1-PkBudg650 background dynamics. This third pathway represents the decarbonization trajectory required to reach absolute sustainability in 2050 for climate change, based on the most stringent SOS threshold (grandfathering for cheese and economic allocation for potato). The modeling approach is detailed in Section 3.6.3.

For both products, the learning-based trajectory remains above the absolute sustainability pathway over the entire assessment period. This confirms that the decarbonization rate resulting from historical learning combined with background improvements is insufficient to remain within the allocated SOS for climate change.

However, in the short term (up to 2030 for cheese, and 2035 for potato), the scenario-based pathway lies below the absolute sustainability pathway. This suggests that the implemented mitigation measures are initially sufficient to meet the required decarbonization rate. However, because these foreground interventions are modelled as time-independent, the resulting reductions remain constant over time. As a consequence, the scenario-based pathway gradually diverges from the SOS trajectory and eventually exceeds it, although it remains closer than the learning-based pathway.

To compare the decarbonization rates required to achieve absolute sustainability by 2050 with those obtained under the scenario-based and learning-based pathways, equivalent Wright learning rates are estimated for each pathway (see Section 3.6.3 for further details). Table 4.7 summarizes the results. These learning rates represent the magnitude of improvement required in foreground systems beyond the changes occurring in the background system under the SSP1-PkBudg650 scenario.

Overall, achieving a climate change impact lower than the allocated SOS by 2050 would require substantially higher learning rates than those observed in either the scenario-based or learning-based approaches. For cheese, a minimum learning rate of 88.7% is required under the grandfathering allocation, while the most optimistic prospective ap-

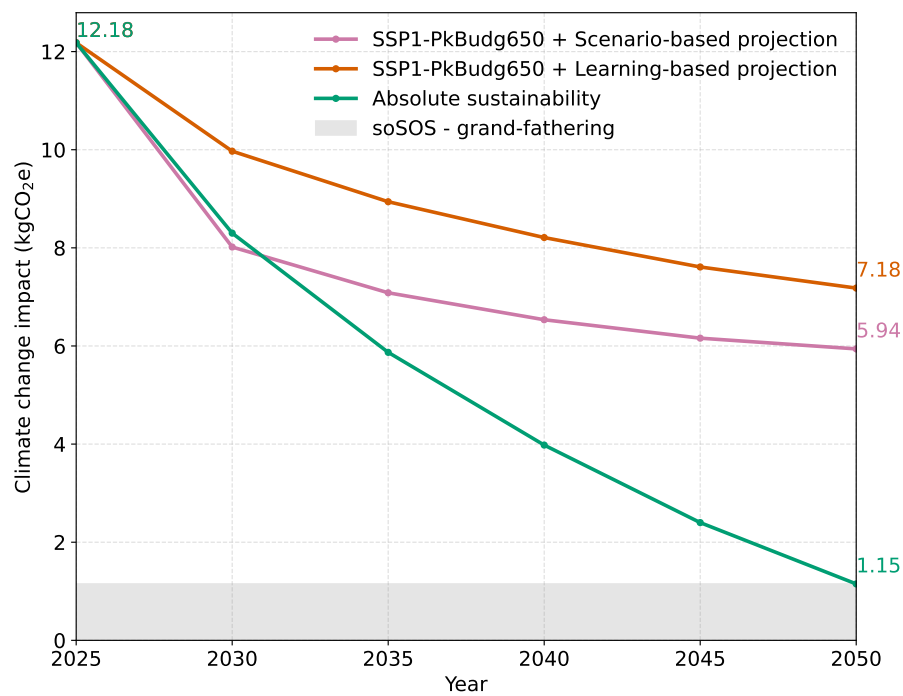


Figure 4.16: Evolution of climate change impacts between 2025 and 2050 under prospective scenarios for cheese. The grey area indicates the range corresponding to the least stringent allocated share of the SOS.

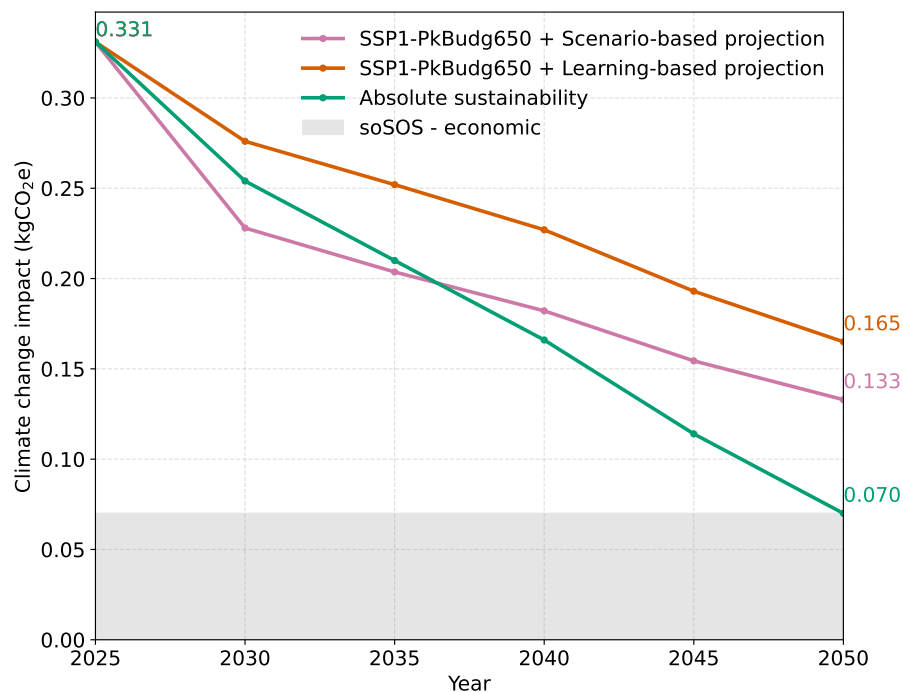


Figure 4.17: Evolution of climate change impacts between 2025 and 2050 under prospective scenarios for potato. The grey area indicates the range corresponding to the least stringent allocated share of the SOS.

Product	Projection pathway	Equivalent learning rate
Cheese	SSP1-PkBudg650 + Learning-based projection	16.2%
	SSP1-PkBudg650 + Scenario-based projection	38.3%
	Absolute sustainability (economic allocation)	92.2%
	Absolute sustainability (grandfathering)	88.7%
Potato	SSP1-PkBudg650 + Learning-based projection	3.5%
	SSP1-PkBudg650 + Scenario-based projection	17.4%
	Absolute sustainability (economic allocation)	42.6%
	Absolute sustainability (grandfathering)	56.5%

Table 4.7: Equivalent learning rates for cheese and potato production systems required to achieve the 2050 impact levels under the different decarbonization pathways, accounting for background evolution under the SSP1-PkBudg650 scenario.

proach achieves only 38.3%. For potatoes, a required learning rate of 42.6% (economic allocation) contrasts with the 17.4% achieved in the scenario-based pathway.

4.6 Sensitivity analysis

4.6.1 Influence of IAM scenario selection on background decarbonization outcomes

To assess the sensitivity of the results to the choice of IAM scenario, an alternative background scenario to the one used in this study (*REMIND – SSP1-PkBudg650*) is considered, namely *IMAGE – SSP1-VLLO*. Figure 4.18 presents the evolution of the climate change impacts for potato and cheese under both scenarios between 2025 and 2050. The baseline impacts, assumed to represent the 2025 situation, are also displayed (black star) to provide context regarding the magnitude of decarbonization assumed in these scenarios compared to the current LCI in *ecoinvent*.

Across all time steps, climate change impacts are higher under *IMAGE – SSP1-VLLO* than under *REMIND – SSP1-PkBudg650*. In 2050, the climate change impact reaches 8.76 kg CO₂e for cheese (+10.1% relative to *REMIND*) and 0.238 kg CO₂e for potato (+37.6% relative to *REMIND*).

This difference in scenario assumptions affects prospective climate change impacts for both products, primarily because the background system decarbonises at a slower rate under *IMAGE*-based pathways. As a consequence, under the learning-based approach, the prospective impacts in 2050 amount to 7.98 kg CO₂e for cheese and 0.231 kg CO₂e for potato, corresponding to increases of 11.2% and 39.9%, respectively, compared to the situation where *SSP1-PkBudg650* is used as the background scenario. Due to time constraints, the scenario-based prospective impacts were not recalculated using the *IMAGE* background scenario. Overall, these results indicate that prospective climate change impacts are sensitive to the choice of IAM, with non-negligible differences observed between products.

4.6.2 Influence of system boundary definition on learning-based projections

To assess the sensitivity of the results to the boundary over which learning effects are applied in the learning-based prospective approach, two alternative scenarios are considered:

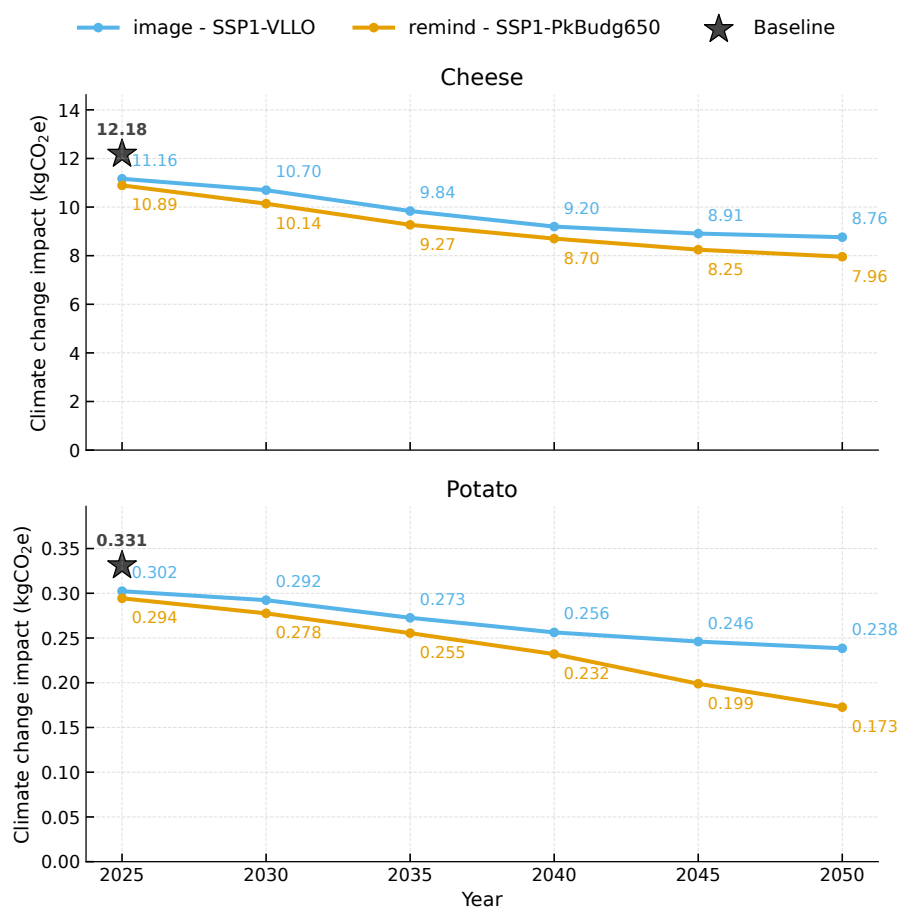


Figure 4.18: Evolution of the climate change impacts between 2025 and 2050 under various background scenarios.

(A) application of the learning rate to first-tier inputs of milk and potato production (e.g. feed crops for cows, tillage operations), in addition to direct emissions from milk and potato production;

(B) application of the learning rate to the entire cheese and potato systems.

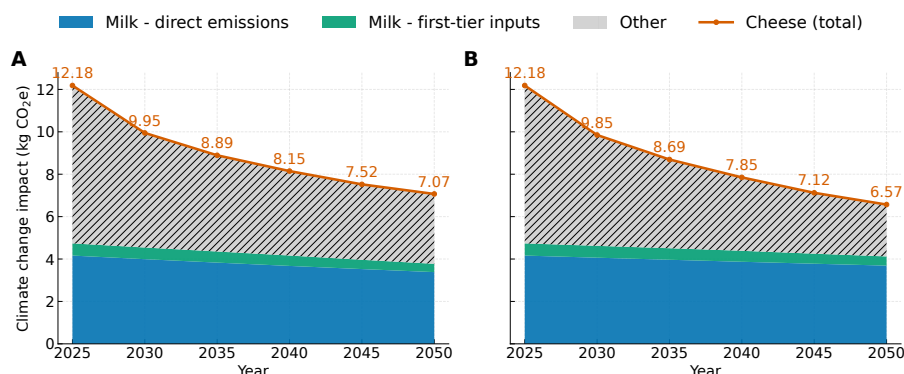


Figure 4.19: Learning-based prospective climate change impacts between 2025 and 2050 for cheese under different boundary for learning application (A: direct+first-tier inputs, B: full system)

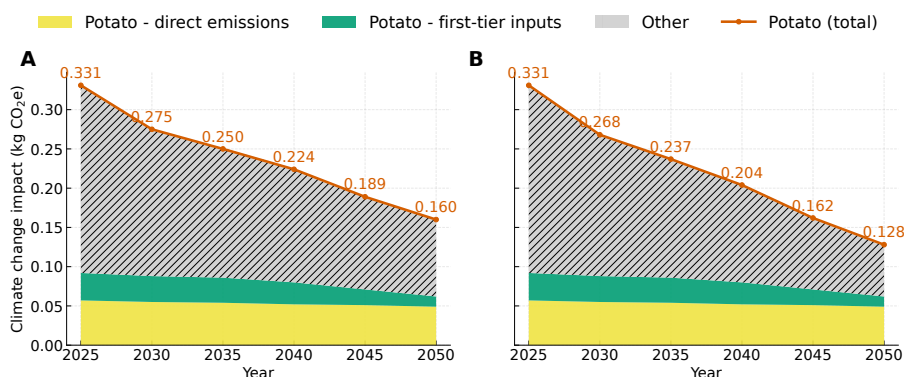


Figure 4.20: Learning-based prospective climate change impacts between 2025 and 2050 for potato under different boundary for learning application (A: direct+first-tier inputs, B: full system)

When learning is additionally applied to first-tier inputs, the total climate change impact further decreases to 7.07 kg CO₂e for cheese (corresponding to an additional 1.5% reduction relative to the reference learning boundary, i.e. when learning is applied only to direct emissions) and to 0.160 kg CO₂e for potato (an additional 2.9% reduction). Under this extended learning assumption, emissions from first-tier inputs decrease by 32% in 2050 compared to the baseline for cheese (versus 13% without learning) and by 62% for potato (versus 49% without learning).

When learning is applied to the full system, the climate change impacts further decrease to 6.57 kg CO₂e for cheese and 0.128 kg CO₂e for potato. This highlights the strong sensitivity of the results to the definition of the system boundary over which learning effects are applied.

5 Discussion

The main results are summarized below and further discussed in the subsequent sections.

First, prospective background mitigation measures result in a reduction of climate change impacts of 34.7% for cheese and 47.8% for potato compared with the baseline situation. However, these reductions represent only one plausible future pathway and remain highly sensitive to the assumptions underlying the IAM scenario. Limitations and recommendations for future studies regarding the use of IAM-based prospective LCI databases are discussed in Section 5.1. Then, additional impact reductions are achieved through mitigation measures and learning effects specific to cheese and potato production systems, resulting in a further reduction of 25.3% for cheese and 23.0% for potato under the scenario-based approach. The challenges and opportunities related to extending the proposed methods and findings to other food products are discussed in Section 5.2.

Absolute sustainability exceedance factors vary substantially depending on the allocation method, ranging from 4.7 to 10.7 for baseline potato production and from 10.6 to 26.5 for baseline cheese production. The allocation of shares of the SOS in prospective LCAs is discussed in Section 5.3. Overall, even under the most optimistic mitigation assumptions considered in this study, absolute sustainability is not achieved for either product. This finding highlights the importance of demand-side mitigation measures as a complement to production-side improvements. Such measures are discussed in Section 5.4. More general limitations of the study and recommendations for future research are presented in Section 5.5.

5.1 Implications of IAM-based prospective LCI databases

When assessing the prospective life cycle impacts of products over the long term, modifying LCI databases to better reflect future environmental impacts and emerging technologies is essential. *premise* is one of the most widely used tools for projecting LCI databases into the future and has gained significant momentum since its publication in 2022. However, several studies call for a more responsible use of IAM-based prospective databases, such as those generated by *premise*, due to a number of methodological limitations, and provide corresponding recommendations [50, 51]. These recommendations are discussed in relation to the present case study.

First, scenarios embedded in IAMs are primarily designed to support climate change mitigation pathways [52]. As a result, prospective databases generated under scenarios such as REMIND-SSP1-PkBudg650 should be used with caution when interpreting impact categories beyond climate change. In particular, burden-shifting effects described in section 4.2.3 are subject to significant uncertainty. This uncertainty stems from the limited representation in IAMs of the effects of mitigation measures on non-climate impact categories, as well as from the lack of an explicit representation of the trade-offs associated with climate change mitigation strategies. For instance, ozone depletion is one of the impact categories that exhibits the most pronounced burden-shifting effects in this study. These prospective results may be overestimated, as potential future bans on ozone-depleting substances are not explicitly represented in IAM scenarios.

Moreover, the LCIA method used in this study (EF 3.1) does not include prospective characterization factors, which further increases the uncertainty associated with impact assessment in prospective LCA.

One of the main advantages of using IAM-based prospective databases in this study lies in its broad sectoral coverage, enabling the assessment of mitigation potentials based on a global transformation of the economy. However, some sectors lack resolution and technological detail, such as the chemical industry, which is often aggregated within broader industrial categories. In this context, approaches such as *premise* community scenarios are particularly relevant for future studies on food systems [53]. They provide sector-specific scenarios derived from the literature with a higher level of detail (e.g., ammonia production pathways) and could be used to refine the prospective modeling of major climate change impact hotspots, such as ammonia production.

IAM-based prospective LCI databases are strongly influenced by the underlying IAM assumptions and the choice of scenario. Therefore, results should be interpreted as representing one plausible future among many, rather than as predictive outcomes guaranteeing future environmental improvements. Such reductions rely on assumptions of substantial technological change and generally do not account for alternative socio-economic pathways, including sufficiency strategies or lifestyle changes, which may be necessary to complement technological transitions. Future work should build on the sensitivity analysis conducted in this study and incorporate a broader range of background scenarios in order to provide a more comprehensive representation of possible futures.

Even though IAM assumptions are crucial to best interpret prospective LCA results, it remains challenging to understand in detail the implications of specific mitigation measures implemented within IAMs on the background modeling in this case study. Some insights were obtained from the analysis of background improvements in relation to production mixes and technological substitutions introduced by IAMs (Section 4.2.1), but a more detailed interpretation remains difficult. Alternative approaches for generating prospective databases are emerging to overcome this limitation.

Brightway-based frameworks such as Futura and Fossil-free *ecoinvent* aim to address this challenge by providing LCA practitioners with greater control over scenario assumptions [54, 55]. The Fossil-free *ecoinvent* framework focuses on modifying fossil-based flows across sectors such as transport and electricity generation using a set of 48 user-defined parameters, including the share of fossil electricity to be decarbonised and the powertrain mix in freight transport. Futura is an interface that allows practitioners to modify market mixes or introduce new technology datasets based on user-defined scenarios, thereby increasing transparency and flexibility in prospective LCA modeling. Such frameworks can be used in future studies to better assess the sensitivity of results to background system improvements.

5.2 Transferability of methods and results to other food systems

The study focuses on cheese and potato products but should be extended to other food products in order to assess prospective environmental impacts at the diet level. The transferability of both the results and the modeling approaches used in this study to other food products is discussed in this section.

Firstly, the background modeling is not product-dependent, as the same IAM scenario is applied to the entire *ecoinvent* database. However, extending this approach to a broader range of food products is limited by the relatively small number of food-specific processes in *ecoinvent*. More detailed LCI databases specific to the agri-food sector, such as Agribalyse, are therefore necessary to broaden the scope of the analysis [56]. While enabling a more detailed representation of food systems, the integration of such sector-

specific databases comes with technical limitations. First, the most recent versions of Agribalyse are not officially integrated into Brightway. Second, the *premise* framework is designed to transform only *ecoinvent* processes. Although Agribalyse relies on *ecoinvent* for part of its background modeling, adapting *premise* to modify Agribalyse-based inventories would require substantial code modifications and further Python development.

Regarding the foreground modeling, most of the agri-food scenarios used in this study can be transferred to other food products. For instance, enteric fermentation mitigation scenarios could be applied to ruminant meat and all dairy products beyond cheese. Similarly, scenarios involving changes in agricultural practices or the use of enhanced-efficiency fertilizers are not specific to potato cultivation. The same set of mitigation measures can therefore be applied to other crop products, consistent with the approach used in this study, where scenarios are applied to feed crops for dairy systems and potatoes. These scenarios provide an initial representation of prospective changes in the foreground system, which could be further refined by incorporating more product-specific mitigation options, such as methane reduction strategies for rice cultivation systems.

Regarding the learning-based approach, the same formulations used in this study can be applied to other food products, provided that product-specific learning rates are available. The FAO dataset used in this study can be leveraged to estimate historical learning rates in other food systems such as eggs or meat production. However, beyond these systems, estimating learning rates would require the use of proxies or datasets with lower data quality. Alternatively, tabulated learning rate values based on the maturity level of the food production system could be used as approximations, for instance using the dataset proposed by Thomassen et al. [57].

5.3 Implications of SOS allocation in AESA

In this study, two allocation methods for the SOS related to climate change are compared: economic allocation and grandfathering. Both methods have limitations when applied to prospective AESA.

First, economic allocation is sensitive to fluctuations in market prices that may result from economic conditions rather than changes in environmental performance. As a result, the allocated share of the SOS may increase under inflation without any corresponding change in environmental impacts. Furthermore, this approach allocates a larger share of the SOS to higher-value products, implicitly assigning greater importance to luxury goods such as cheese than to staple foods such as potatoes. In contrast, the grandfathering approach reflects the current distribution of environmental impacts and therefore tends to favour products with the highest historical emissions by allocating them a larger share of the SOS. This creates a status quo in which highly emitting products are granted a greater environmental budget despite their larger contribution to environmental pressures.

Moreover, although the allocated shares of the SOS are assumed to remain constant over time in this study, the allocation of planetary boundaries is inherently time-dependent in the context of prospective LCA. Indeed, the resulting absolute sustainability thresholds derived from both economic and grandfathering approaches rely on time-dependent parameters. For instance, in an economic allocation framework, variables such as prices and final consumption expenditures evolve over time. Consequently, the share of the SOS allocated to food products would also vary over time as the relative economic weight of these products within total consumption changes.

More generally, neither of these product-level allocation methods guarantees consistency with a sustainable diet. Alternative approaches are emerging in which the environmental

budget allocated to the food sector is further downscaled to food groups or individual food products while ensuring that aggregate food consumption remains within planetary limits [58]. In such approaches, a larger share of the budget is typically allocated to foods with lower environmental impacts, such as plant-based products, and a smaller share to more emission-intensive foods.

5.4 Role of demand-side mitigation in achieving absolute sustainability

This study assesses climate change impacts at the product level from an emission intensity perspective looking at the reductions achievable by only acting on the production-side. One of the main takeaways of this study is that improvements on the production side alone are insufficient to remain within absolute sustainability thresholds by 2050. Even under the most optimistic mitigation assumptions considered in this study, exceedance ratios reach values above 1.9 for potato and 5.2 for cheese.

Moreover, these results are derived under a highly optimistic assumption regarding technological progress, based on the SSP1-PkBudg650 scenario, which reaches approximately +1.37°C of GMST increase in 2100 relative to pre-industrial levels. In contrast, warming under current policies is projected to reach a median value of approximately +2.7°C, according to estimates from Hausfather [59]. This suggests that exceedance ratios would likely be substantially higher in 2050 under real-world policy trajectories than those obtained in this study.

Based on this case study, these findings provide initial insights suggesting that the food system cannot remain within planetary boundary constraints for climate change by relying solely on reductions in emission intensities. Sufficiency-based mitigation measures are therefore also required on the demand side, in particular through a reduction in the consumption of emission-intensive products. In the context of this study, the baseline climate change impact of cheese is approximately 37 times higher than that of potato per kilogram of product, increasing to 45 times higher in 2050 even after the implementation of production-side mitigation measures. As a result, reducing cheese consumption appears necessary to approach compliance with planetary boundary thresholds at the dietary level.

Recommendations for such dietary shifts are provided in reference studies such as the EAT-Lancet Commission, which proposes dietary patterns that deliver both health and environmental benefits. These shifts include increased consumption of fruits, vegetables, legumes, nuts, and whole grains, alongside a substantial reduction in the consumption of red meat. The intake of poultry, fish, and dairy products should remain moderate, depending on cultural and nutritional contexts [2].

Extending the results of the case study to the dietary level and considering an allocation of the SOS at this scale would complement these findings by identifying the dietary changes required to comply with planetary boundaries under assumptions of prospective reductions in emission intensities driven by production-side mitigation measures.

5.5 Limitations and suggestions for further studies

Some limitations of this study, as well as suggestions for future improvements, are provided below.

- **Baseline inventory representativeness** The learning-based pLCA approach required the selection of a reference year for the system, which was set to 2025 in

this study. However, *ecoinvent* inventories are not fully representative of present-day conditions. For instance, only milk production datasets for Canada (Quebec) and South Africa are available for the cheese process, although these countries accounted for only 1.67% of global cow milk production in 2024 [60]. Furthermore, the underlying milk production datasets are based on data from 2010 and 2016, while the yield data used in potato production processes correspond to the 2001-2006 period.

This limitation could be addressed in future studies by updating key parameters, such as milk emission intensities, or by scaling agricultural yields to current values in order to improve representativeness under present-day conditions. The background inventory could also be made more representative of current conditions by relying on prospective databases based on IAM scenarios calibrated to the year 2025.

- **Temporal dynamics** Under the scenario-based approach, foreground modeling relies on agri-food mitigation scenarios that are parameterized identically across all time steps from 2030 to 2050. While these scenarios are representative of mitigation measures and associated effects currently documented in the literature, future technological progress may enable greater impact reductions than those currently reported, and additional mitigation measures may emerge over time. Scenario modeling could therefore be improved by incorporating an evolving set of mitigation measures and by allowing their associated reduction potentials to change over time. In the learning-based approach, the projected reduction in climate change impacts may be underestimated. Indeed, the learning rate used in this study is an aggregate parameter representing all learning mechanisms correlated with cumulative production. However, as the projected growth in production is relatively low for both products, learning effects are also projected to remain limited. This may underrepresent certain innovation mechanisms, such as R&D-driven or time-dependent technological progress, which are not strictly tied to production levels and may instead correlate with variables such as time or investment levels.
- **Mitigation potential uncertainty** As this study relies on potato and cheese processes already modeled in *ecoinvent*, limited product-specific information is available (e.g., cattle breed, soil conditions, or climatic context). Consequently, mitigation measures are modelled using generic milk and crop production systems derived from meta-analyses. The findings should be interpreted with caution when applied to specific production systems, as local conditions may significantly affect the prospective climate change results. This is particularly relevant for potato, as generic crop mitigation scenarios were used in the scenario-based approach, and cereals were used as a proxy for learning effects in the learning-based approach. Further studies should therefore integrate more product-specific literature to improve the representativeness of the results.
- **Nitrogen cycle modeling** Limited information is provided in *ecoinvent* regarding the modeling of nitrogen cycling in the potato system, particularly the relationship between nitrogen fertilizer inputs and N_2O emissions. For the fertilizer reduction (FR) scenario, it is therefore not possible to explicitly track the impact of reduced nitrogen inputs on all nitrogen-related biosphere flows (e.g., N_2O , NH_3). Instead, the reduction in N_2O emissions associated with this mitigation measure is applied exogenously, which partially decouples nitrogen inputs and outputs and simplifies the underlying biogeochemical processes.
- **Agricultural CO_2 emissions** The lack of consistent information on CO_2 emission

reductions from direct agricultural processes introduces uncertainty in scenario comparisons. CO₂ emissions reductions are only explicitly reported for nitrification inhibitor (NI) scenario based on Chen et al., while no consistent data are available for other mitigation measures [38]. To improve consistency, either harmonised data on CO₂ reductions in agricultural systems should be used, or these effects should be excluded for all scenarios.

- **Pathway completeness** Both decarbonization pathways could be further refined by incorporating additional mitigation measures. For instance, in the learning-based pathway, more detailed learning rates could be introduced for cheese processing, reflecting potential efficiency gains from accumulated production experience. Similarly, agri-food scenarios could integrate additional non-technological measures including productivity improvements.
- **Learning-based approach** In the learning-based approach, emission reductions in the foreground system are estimated independently of changes occurring in the background system. As a result, applying learning rates at the system level may lead to unrealistic extrapolations, for instance when projected foreground improvements exceed the magnitude of residual emissions from the background system. Future studies should follow the approach adopted in the main results of this work and apply learning rates with caution, restricting their use to system components for which technological or biological improvements can reasonably be attributed to observable learning processes. Learning rates should not be used as a generic proxy for undifferentiated technological progress across an entire system.
- **Computational constraints** Generating prospective background databases using *premise* for a given year is computationally demanding and required the use of high-performance computing resources. Database generation can take up to three hours per scenario per year, limiting the feasibility of exploring a large number of scenarios. To improve computational efficiency and reduce storage requirements, the so-called “superstructure” approach can be used [22]. In this approach, a single modified *ecoinvent* database is stored in Brightway per scenario, while differences in process flows across years are summarized in a single SDF that can be imported into AB to generate prospective results for a given year. Although more efficient than the generation of separate databases for each time step used in this study, the superstructure approach is not compatible with the foreground modeling framework adopted here. This is because the foreground modeling requires a time-specific prospective *ecoinvent* database to be available in Brightway for each time step in order to construct the corresponding SDF. This requirement is not fulfilled by the superstructure approach, where time-specific background databases are not stored independently but generated dynamically during impact assessment. Further research should therefore focus on adapting foreground modeling to enable compatibility with the superstructure approach, or on exploring alternative methods for integrating background system changes into prospective LCA.

6 Conclusion

The objectives of this study were to estimate the prospective climate change impacts of cheese and potato under production-side mitigation strategies, and to assess the potential for achieving absolute sustainability targets by 2050. The results show that substantial impact reductions can be achieved through transformations in both agri-food systems and other economic sectors. In particular, the decarbonization of electricity and heat production, as well as the electrification of freight transport, play a key role in reducing climate change impacts. Within agri-food systems, the most effective mitigation options include the use of enhanced-efficiency fertilizers through nitrification inhibitors and reductions in methane emissions from cattle, while the electrification of agricultural machinery provides more limited benefits. Overall, under an optimistic set of socio-economic conditions and mitigation options, impacts in 2050 could be reduced to 5.94 kg CO₂ eq for cheese (-51.2% relative to the baseline) and 0.133 kg CO₂ eq for potato (-59.8%).

However, these impacts still exceed the absolute sustainability thresholds in 2050 under all allocation methods of the safe operating space for climate change considered in this study, by factors of 5.2 for cheese and 1.9 for potato under the most optimistic scenario. Absolute sustainability in 2050 therefore cannot realistically be achieved for the studied food products through production-side mitigation measures alone. Achieving such outcomes would require the combination of a highly optimistic socio-economic pathway focused on sustainability (such as SSP1-PkBudg650 considered in this study) and environmental learning rates in agri-food systems substantially higher than historical trends.

These results suggest that broader transformations beyond production systems are required to remain within planetary boundaries. In particular, sufficiency-oriented strategies, including shifts towards lower-impact dietary patterns, and reductions in food waste, appear necessary to complement production-side mitigation efforts.

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Appendix

Abbreviations

Abbreviation	Definition
AESA	Absolute Environmental Sustainability Assessment
AD	Agricultural Direct emissions mitigation scenarios
AM	Agricultural Machinery emissions mitigation scenarios
CAGR	Compound Annual Growth Rate
EC	Economic allocation
EF	Environmental Footprint method
ENT	Enteric fermentation mitigation scenarios
EpC	Equal per capita allocation
FAO	Food and Agriculture Organization of the United Nations
FL	Increased Feeding Level mitigation scenario
FP	Optimized Fertilizer Placement mitigation scenario
FR	Optimized Fertilizer Rate mitigation scenario
GF	Grandfathering allocation
GMST	Global Mean Surface Temperature
IAM	Integrated Assessment Model
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
LCIA	Life Cycle Impact Assessment
LHV	Lower Heating Value
LR	Learning Rate
NI	Nitrification Inhibitors mitigation scenario
NI+UI	Double inhibitors mitigation scenario
NOP	3-NOP supplementation mitigation scenario
NS	Nitrate Supplementation mitigation scenario
PB	Planetary Boundary
pLCA	Prospective Life Cycle Assessment
RoW	Rest Of the World
SDF	Scenario Difference File
SOS	Safe Operating Space
soSOS	Share Of Safe Operating Space
SSP	Shared Socioeconomic Pathway
T_d	Doubling time
UI	Urease Inhibitors mitigation scenario
VLLO	Very Low Overshoot

Methodological details

A scenario difference file contains all exchanges for a given set of activities in a Brightway database, where each row represents a single exchange between a source (from) and a receiving (to) activity.

The "from" section specifies the origin of each exchange, including the activity, reference product, location, database, and associated metadata, while the "to" section describes the receiving activity in a consistent manner. The exchange amount is then defined in dedicated columns for one or more scenarios, enabling alternative assumptions to be

Impact category	Unit	Normalization factor	Weighting factor
Acidification	mol H ⁺ eq./person	$5,56 \times 10^1$	6,2%
Climate change	kg CO ₂ eq./person	$7,55 \times 10^3$	21,1%
Ecotoxicity, freshwater	CTUe/person	$5,67 \times 10^4$	1,9%
EF-particulate matter	disease incidences/person	$5,95 \times 10^{-4}$	9,0%
Eutrophication, freshwater	kg P eq./person	$1,61 \times 10^0$	2,8%
Eutrophication, marine	kg N eq./person	$1,95 \times 10^1$	3,0%
Eutrophication, terrestrial	mol N eq./person	$1,77 \times 10^2$	3,7%
Human toxicity, cancer	CTUh/person	$1,73 \times 10^{-5}$	2,1%
Human toxicity, non-cancer	CTUh/person	$1,29 \times 10^{-4}$	1,8%
Ionising radiation	kBq U-235 eq./person	$4,22 \times 10^3$	5,0%
Land use	pt/person	$8,19 \times 10^5$	7,9%
Ozone depletion	kg CFC-11 eq./person	$5,23 \times 10^{-2}$	6,3%
Photochemical ozone formation	kg NMVOC eq./person	$4,09 \times 10^1$	4,8%
Resource depletion, fossils	MJ/person	$6,50 \times 10^4$	8,3%
Resource depletion, minerals and metals	kg Sb eq./person	$6,36 \times 10^{-2}$	7,6%
Water use	m ³ water eq. deprived/person	$1,15 \times 10^4$	8,5%

Table 1: Normalization and weighting factors for endpoint calculations (EF 3.1).

represented through additional scenario-specific columns containing modified values for the same exchange.

Part a) Exchange identification												Part b) Scenario values	
Flow from						Flow to						Scenarios	
from activity name	from reference product	from location	from categories	from database	from key	to activity name	to reference product	to location	to categories	to database	to key	baseline	less emissions
market for natural gas, high pressure Carbon dioxide, fossil	natural gas, high pressure	US		ecoinvent-3.12-cutoff		electricity production, natural gas combined cycle	electricity HV	US-NPCC		ecoinvent-3.12-cutoff		0.183	0.150
			air emissions (high stacks)	biosphere-3		electricity production, natural gas combined cycle	electricity HV	US-NPCC		ecoinvent-3.12-cutoff		0.381	0.350

Table 2: Example of an SDF for a scenario representing a lower-emission gas power plant.

Additional results

Reference product	Dry matter / fresh mass ratio
alfalfa-grass silage	1.00
barley grain, feed	0.86
energy feed, gross	5.10
hay	1.00
maize grain, feed	0.86
maize silage	0.28
oat grain	0.865
protein feed, 100% crude	1.00
soybean, feed	0.89
straw	0.85
wheat grain, feed	0.85
wheat bran	0.87
rape meal	0.90

Table 3: Conversion factors of feed inputs to dry matter based on *ecoinvent*.

Impact category	Unit	Cheese	Potato
Acidification	mol H ⁺ eq	0.131	6.89×10^{-3}
Climate change	kg CO ₂ eq	12.1	0.331
Ecotoxicity, freshwater	CTUe	103	14.9
EF-particulate matter	Disease incidence	1.08×10^{-6}	5.32×10^{-8}
Eutrophication, freshwater	kg P eq	2.46×10^{-3}	2.05×10^{-4}
Eutrophication, marine	kg N eq	0.0544	3.81×10^{-3}
Eutrophication, terrestrial	mol N eq	0.536	0.0293
Human toxicity, cancer	CTUh	4.16×10^{-9}	3.78×10^{-10}
Human toxicity, non-cancer	CTUh	7.43×10^{-8}	2.17×10^{-8}
Ionising radiation	kBq ²³⁵ U eq	0.287	7.24×10^{-3}
Land use	dimensionless	5.47×10^3	23.4
Ozone depletion	kg CFC-11 eq	3.10×10^{-7}	1.62×10^{-8}
Photochemical ozone formation	kg NMVOC eq	0.0326	1.90×10^{-3}
Resource depletion, fossils	MJ	75.4	3.16
Resource depletion, minerals and metals	kg Sb eq	6.08×10^{-5}	2.27×10^{-6}
Water use	m ³ world eq. deprived	12.2	3.41

Table 4: Midpoint environmental impacts of under baseline conditions (EF 3.1).

Cheese		
Impact category	Variation	Main drivers
Ozone depletion	+26.9%	• Methyl chloride (+7300%) • Refrigerant R134a (+825%)
Human toxicity, non-cancer	+18.0%	• Copper anode (+73%) • Steel production (+291%) • Zinc production (+378%)
Human toxicity, cancer	+11.9%	• Ferrochromium (+58%) • Copper anode (+64%) • Nickel sulfate (+10037%)
Potato		
Impact category	Variation	Main drivers
Human toxicity, cancer	+10.7%	• Ferrochromium (+73%) • Copper anode (+100%)
Material resources	+10.6%	• Lead concentrate (+231%) • Silver (+2401%) • Gold (+32%)
Ozone depletion	+7.3%	• Methyl chloride (+4047%) • Refrigerant R134a (+40%) • Purified terephthalic acid (+110%)
Land use	+7.2%	• Logging (+64%)
Human toxicity, non-cancer	+5.9%	• Copper anode (+111%) • Steel production (+358%)

Table 5: Impact categories exhibiting variations exceeding 5% between SSP1-PkBudg650 and baseline conditions, along with the main processes responsible for burden shifting.

Supplementary Materials

SM1 Scenario difference files

SDF files corresponding to the agri-food scenarios described in Section 3.4 are provided as supplementary material. The files are organized in folders by year (2030, 2035, 2040, 2045, and 2050). Scenario difference files are named according to the scenario group and the affected product.

Please note that scenarios targeting direct agricultural emissions (AD) are split into two separate SDF files: one covering scenarios related to agricultural practices (FP and FR), and another covering scenarios related to the use of enhanced-efficiency fertilizers (NI, UI, and NI+UI).

SM2 Calculations and results

An Excel file presenting the calculations and results is provided as supplementary material. The file is organized in four sections : Parameters, Scenario-based LCA, Learning-based LCA and Absolute sustainability. The content of each is described in Table 6 :

Section	Content	Related figures/tables
Parameters	- Parameters used for the calculation of SOS allocation and the projection of historical learning in the learning-based approach. - Forecasted production quantities [7]. - LCIA-consistent planetary boundaries [18].	Table 4.5
Scenario-based LCA	- Individual contribution of activities to impacts under all scenarios. - Summary of impacts by impact category and scenario. - Dashboard identifying hotspots by scenario, year, impact category, and product. - Summary of additional mitigation potential of agri-food scenarios. - Summary of baseline midpoint results.	Figures 4.2a, 4.2b, 4.3, 4.4, 4.8, 4.9, 4.10a, 4.10b, 4.18 Table 4.2, 4.3, Appendix 4
Learning-based LCA	Climate change impacts projected under learning effects.	Figures 4.12, 4.13, 4.15a, 4.15b, 4.19, 4.20
Absolute sustainability	- Calculation of SOS allocation. - Calculation of absolute sustainability curves. - Calculation of equivalent learning rates.	Figures 4.12, 4.13 Table 4.6, 4.7

Table 6: Overview of supplementary file content and corresponding sections in the main report.

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